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Analysis of Characteristics of Vertical-Axis Turbines Using Computational Fluid Dynamics

MINH THAO NGUYEN

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Abstract

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This thesis presents numerical analyses of straight-bladed vertical-axis turbines with respect to turbine performances and dynamic structural forces. To deepen understanding of the turbine operation and productivity, the effects of geometric parameters and operating conditions on turbine outcomes are essential to be investigated.

The turbine configurations are modelled by using two-dimensional computational fluid dynamics (CFD) simulations. The CFD models are verified through grid and time-step independence studies, and are further validated against field experimental data for both the vertical-axis wind and marine current turbines. Overall, the numerical results show a reasonably good agreement with experimental measurements, particularly at medium-to-high tip-speed ratios.

Once the turbine models are validated, the unsteady structural forces of a 12 kW wind turbine are first examined under varying wind speeds and tip-speed ratios. Effects of these operating conditions on the aerodynamic forces are analyzed, and the normal force distributions are compared with open-field measurements to identify physically relevant phenomena during the turbine rotation. This is considered a key contribution of the study, as the scarcity of detailed experimental data of the wind turbine forces often limits the reliability of numerical investigations. Second, a comprehensive analysis is performed on a 7.5 kW marine current turbine to evaluate the influences of different fixed blade-pitch angles. Determining optimal pitch angles is shown to be a critical factor in the development and design of the water turbines. Finally, the combined effects of geometric and operational parameters, including the number of blades, turbine radius, chord length, tip-speed ratio, incoming flow velocity, and Reynolds number, are assessed for hydrokinetic rotors performances.

Collectively, the findings from the aforementioned studies provide valuable insights into the impacts of the design and operational factors on the turbine power extraction and dynamic loading characteristics, thereby forming a solid basis for the optimization of the vertical-axis turbine designs in both the wind and marine current applications.

Keywords: straight-bladed vertical-axis turbine, wind power, marine current power, hydrodynamics, unsteady force, solidity, turbine geometry, operational condition, CFD, URANS

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To my lovely family
Viet Thang, Lynna & August

List of papers

This thesis is based on the following papers, which are referred to in the text by their Roman numerals.

I. **Minh Thao N.**, J. Forslund, K. Thomas, A. Goude (2019).
"Fault analysis of a marine current vertical-axis turbine missing two blades"
13th European Wave and Tidal Energy Conference, Naples, Italy, 1769.

II. **Minh Thao N.**, Balduzzi F., A. Bianchini, G. Ferrara, A. Goude (2020).
"Evaluation of the unsteady aerodynamic forces acting on a vertical-axis turbine by means of numerical simulations and open site experiments"
Journal of Wind Engineering and Industrial Aerodynamics, vol. 198, 104093.
https://doi.org/10.1016/j.jweia.2020.104093.

III. **Minh Thao N.**, F. Balduzzi, A. Goude (2021).
"Effect of pitch angle on power and hydrodynamics of a vertical axis turbine"
Journal of Ocean Engineering, vol. 238, 109335.
https://doi.org/10.1016/j.oceaneng.2021.109335.

IV. **Minh Thao N.**, F. Balduzzi, A. Bianchini, A. Goude (2026).
"Assessing the influence of operational and geometric parameters on the hydrodynamic performance of vertical-axis marine current turbines"
Journal of Marine Science and Application, accepted.
https://doi.org/10.1007/s11804-026-00911-9.

V. C. Bittencourt Sales Junior, **Minh Thao N.**, F. Balduzzi, A. Goude, A. Sarmiento (2026).
"Integrated evaluation of geometric and operating parameters on H-Darrieus hydrodynamic turbine performance and structural response", *Journal of Marine Science and Application,* submitted.

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The author has also contributed to the following paper which is not included in this thesis.

Minh Thao N., I. Dolguntseva, M. Leijon (2015).
"CFD analysis on the effect of the tidal grating in opposite directions", *XIII International Conference on Marine Science and Technology,* Boston, USA.

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Nomenclature

Symbol	SI-unit	Description
A_b	m^2	Total blade area
A_c	m^2	Total channel area
A_s	m^2	Rotor swept area
c	m	Blade chord
C_D	-	Drag coefficient
C_L	-	Lift coefficient
C_P	-	Turbine power coefficient
C_{P_max}	-	Maximum turbine power coefficient
D	m	Rotor diameter
D_{2D}	N/m	Two-dimensional drag
d_i	m	Domain length from the inlet to the rotor centre
d_o	m	Domain length from the rotor centre to the outlet
F_n, F_t	N	Normal and tangential force of a blade
F_{xi}, F_{yi}	N/m	Vertical and horizontal force of a blade
H	m	Blade height
i	-	Index of blade numbers
L_{2D}	N/m	Two-dimensional lift
N	-	Number of blades
$P_{turbine}$	W	Net turbine power
P_{loss}	W	System losses
R	m	Rotor radius
Re	-	Reynolds number
Re_c	-	Chord-based Reynolds number
T_{av}	Nm	Average torque acting on turbine
T_{res}	Nm	Experimental resistant torque
y+	-	Dimensionless wall distance
V	m/s	Undisturbed incoming velocity
V_R	m/s	Rotational velocity of a turbine

V_∞	m/s	Incoming velocity
W	m/s	Relative velocity of a turbine
ρ	kg/m^3	Density of fluid
υ	m^2/s	Fluid kinematic viscosity
Ω	rad/s	Angular velocity
θ	deg	Azimuthal angle
β	deg	Pitch angle
ϵ	-	Blockage ratio

Abbreviations

Abbreviations	Description
2D	Two-dimensional
3D	Three-dimensional
ALM	Actuator line model
BEM	Blade element momentum model
CFD	Computational fluid dynamics
HATs	Horizontal-axis turbines
RANS	Reynolds-averaged Navier-Stokes
SST	Shear stress transport
TSR	Tip-speed ratio
TSR_{opt}	Optimal tip-speed ratio
URANS	Unsteady Reynolds-averaged Navier-Stokes
VAMTs	Vertical-axis marine current turbines
VATs	Vertical-axis turbines
VAWTs	Vertical-axis wind turbines

1. Introduction

This thesis investigates the fluid dynamics, structural loading, and performance of vertical-axis turbines (VATs). The study focuses on medium-scale turbines, a range that has received limited attention in prior research due to mainly a lack of experimental data to validate the numerical models and therefore represents a notable gap in current knowledge.

First, the focus is on the wind application, where the unsteady structural forces acting on the turbine are analyzed in detail to improve understanding of their dynamic behavior. Second, for the marine current application, the combined influence of geometric parameters and operational conditions on turbine loading and power performance is systematically examined.

The analyses are conducted using high-fidelity numerical simulations based on computational fluid dynamics (CFD). The numerical models are validated against the field experimental data to ensure reliability and to strengthen the relevance of the findings for practical turbine design and performance prediction.

1.1 Background

Global greenhouse gas emissions continue to escalate, primarily driven by unsustainable patterns of energy consumption, land use, production, and consumption practices across regions, nations, and individuals [1,2]. To mitigate this increase necessitates coordinated global actions [3]. The European Union has pledged to become the first climate-neutral continent, targeting net-zero greenhouse gas emissions by 2050 [4]. Similarly, China has committed to achieving carbon neutrality by 2060 [5], while India has set a goal of reaching net-zero emissions by 2070 [6,7]. The attainment of carbon neutrality is central to fulfil the targets established by the Intergovernmental Panel on Climate Change and constitutes a core objective of the United Nations Sustainable Development Goals [8]. In pursuit of those goals, numerous policy measures have been introduced to accelerate renewable energy deployment [7,8]. Given the pivotal role of renewable energy in decarbonisation, the expansion and advancement of clean, sustainable energy technologies is of urgent priority. Global demand for renewable sources such as wind, solar, geothermal, and ocean-based energy is increasing rapidly.

Covering approximately 71% of the Earth's surface, the ocean represents a vast and largely untapped source of the renewable energy [9]. The marine energy resources can be harnessed primarily through wave and marine current technologies. At present, the marine current energy is transitioning from pilot-scale projects to pre-commercial deployments, while the wave energy technologies remain at earlier stages, progressing from prototype testing toward pilot-scale applications [10–12]. In contrast, wind power has already established itself as the most technologically mature, commercially viable, and widely deployed technology, acknowledged by both academia and industry for its scalability, resource abundance, and potential to deliver the clean energy at scale. Notably, in 2024 alone, a record with a power of 117 GW of new wind energy capacity was installed worldwide [13].

Wind energy is typically harvested through wind turbines, whereas tidal current energy can be harnessed using underwater turbines, barrages, or alternative devices such as venturi systems and oscillating foils [14]. Both the wind and marine current turbines operate on a similar fundamental working principle: the conversion of the kinetic energy of moving air or water into rotational mechanical energy, which is subsequently transformed into electrical energy via a generator [15,16]. In spite of the same operational principle, the two technologies differ in several important respects:

- **Predictability and reliability**: tidal current energy is inherently more predictable and consistent than wind energy, as it is driven by the regular gravitational interactions between the Earth, Moon, and Sun, resulting in consistent and foreseeable flow patterns.
- **Turbine size**: For the same given rated power output, the marine current turbines are generally smaller than wind turbines due to the higher density of water, which enables greater energy extraction at lower flow velocities. There can however be a size limitation to the turbine due to limited water depth.
- **Structural loading**: the marine current turbines experience significantly higher structural loads than the wind turbines due to the higher fluid density of water. Consequently, these water turbine systems require strong blades and support structures for a similar turbine size. There is also a risk of cavitation for high water speeds, which could damage the blades.
- **Environmental challenges**: the marine current turbine systems operate in harsh marine conditions, where they are exposed to challenges such as corrosion and biofouling caused by marine growth, which can affect turbines' performance and durability.

The number of published studies on the straight-bladed VATs remains relatively limited. A literature survey is conducted using Elsevier's Scopus database indicates that the studies focusing on the straight-bladed VATs design

account for only a small proportion of the total VAT-related publications. In particular, the studies on the straight-bladed design are approximately 10% of the total, highlighting a notable imbalance in research attention as illustrated in Figure 1.1.

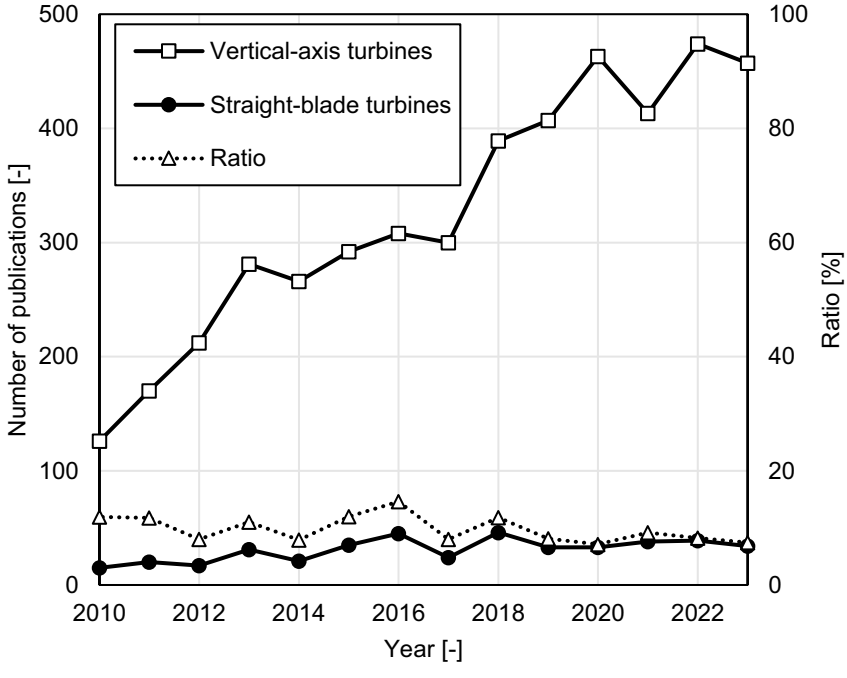


Figure 1.1. Number of publications in VATs from 20210 to 2023.

1.2 Vertical-axis turbine conversion technology

Conversion technologies for wind and tidal current energy are generally categorized into two main configurations based on the orientation of the rotor axis, namely horizontal-axis turbines (HATs) and vertical-axis turbines (VATs). HATs, particularly within wind energy applications, represent a technologically mature and extensively validated design, as evidenced by their widespread deployment in both onshore and offshore environments [13,14,17,18]. However, VATs exhibit several intrinsic structural and operational advantages over HATs, despite their comparatively lower level of technological maturity. In the turbine structure, VATs benefit from a simplified design, as their blades are often characterized by uniform, untwisted cross-sections, which not only streamline the manufacturing process but also enhance feasibility for large-scale production. In the operational perspective, VATs possess the ability to

harness energy from multidirectional flow regimes without requiring active yaw mechanisms, thereby rendering them particularly suitable for environments with variable or turbulent flow conditions. Moreover, VATs provide notable techno-economic benefits in terms of installations and maintenance, as critical heavy components such as the generator can be positioned at the baseline of the system, which can reduce maintenance complexities and enhance operational efficiency. For the onshore wind installations, the generator can be placed at the ground level, therefore, it simplifies installation, inspection, and maintenance due to easier access to components near the ground [19]. Similarly, in the offshore applications, the generator can be put at the waterline on floating platforms [20] or integrated at the seabed level [21]. Overall, these structural and functional advantages underscore the potential of VATs as a promising and scalable technology, with significant implications for future developments in diverse renewable energy sectors.

1.3 Vertical-axis turbine research at Uppsala University

The aforementioned advantages of the vertical-axis turbines (VATs) have been a central motivation for the sustained research activities at the Division of Electricity, Uppsala University, which has been engaged in the development of the VAT-based energy conversion systems since 2002. Within the scope of this long-term initiative, four wind VAT prototypes have been constructed and deployed: an initial 1.5-kW system, a 12-kW prototype, a 10-kW turbine developed for the telecommunications company Ericsson Sweden, and a 200-kW system manufactured by the spin-off company Vertical Wind AB [22–26]. The development is followed by the marine current VAT with a 7.5 kW rated power deployed in a river site.

Extensive investigations of the VAT system have been carried out by both former and current team members. The specific topics include the generator design and optimization [22,25,27], electrical system integration [28–31], resource and site assessments [32], and comprehensive measurement campaigns [23,30,33]. Furthermore, research into the turbine fluid dynamics has yielded substantial contributions to several doctoral theses [34–37]. Their studies have focused on the development of advanced numerical methodologies and theoretical frameworks, including vortex-based models, blade element momentum theory, and actuator line modelling approaches, aimed at analyzing turbine performance, structural loading, flow field characteristics, and aeroacoustics behavior.

Among those forementioned prototypes, the 12 kW wind turbine and 7.5 kW water turbine have been a primary focus in this thesis, and measurements from these turbines are used to validate the numerical turbine models. These two turbine systems differ in the number of blades: the wind turbine features

a three-bladed rotor, while the marine current turbine is equipped with a five-bladed configuration, as depicted in Figure 1.2. The wind VAT was deployed at the Marsta site (coordinates: N 59° 55′ 32″, E 17° 35′ 12″) in 2006, and the water VAT was installed in 2013 in the river Dalälven (coordinates: N 60° 23′ 15.252″, E 17° 14′ 55.212″).

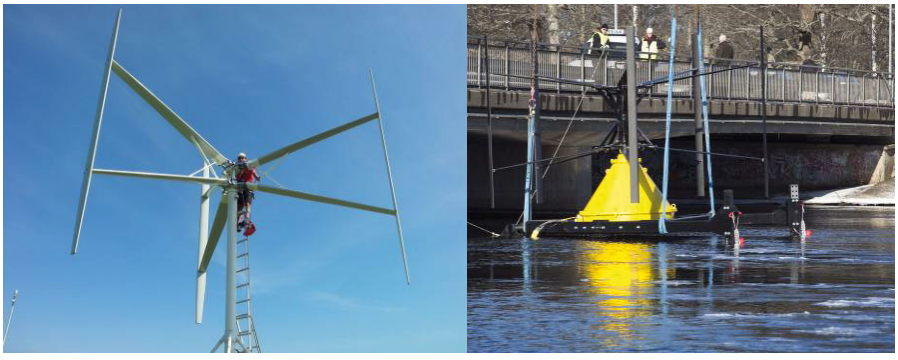


Figure 1.2. Three-blade wind turbine at Marsta site (left), and five-blade marine current turbine at the river Dalälven (right). Photos from Uppsala University.

1.4 Research questions and aims

Despite the significant research interest in vertical-axis turbines, several challenges persist in their development, which continue to represent important research gaps, as outlined below.

Simulation tools: Common approaches, including streamtube, vortex, and actuator line models, are widely used to simulate vertical-axis turbine configurations [35]. These methods rely on simplifying assumptions that reduce computational cost; however, they depend on externally supplied fluid dynamic data, such as lift and drag coefficients, which may limit their predictive accuracy under complex and highly unsteady flow conditions.

Turbine size: The majority of existing studies on vertical-axis wind turbines are conducted on small-scale configurations, generally with rotor radius up to approximately 2 m [38]. This is largely due to limitations in experimental facilities, which constrain testing to laboratory-scale turbines and limit the investigation of full-scale effects.

Operational range: Most studies focus on relatively low Reynolds numbers, typically up to a chord-based Reynolds number of approximately 10^5, where turbine performance is highly sensitive to Reynolds number effects [39]. Consequently, flow phenomena associated with higher Reynolds number regimes are often not fully captured, leading to an incomplete representation of aerodynamic behavior under realistic operating conditions.

Validation: A further critical research gap is the limited availability of field-scale validation data for turbine systems. This includes measurements of structural loads in wind turbines and power coefficients in marine current turbines [40,41]. In tidal current applications, underwater deployment and long-term monitoring introduce significant technical and operational challenges. For example, reported operational failures, such as missing of a blade in experimental marine turbine studies, highlight the demanding nature of field testing [42]. The scarcity of high-quality field data constrains the validation of numerical models and hinders the reliable scaling of turbine designs from laboratory to real-world applications.

This thesis aims to bridge critical research gaps in the development of vertical-axis turbines by (i) using simulation tools capable of accurately capturing unsteady flow behavior without reliance on external fluid dynamic inputs, (ii) examining turbine performance and fluid dynamic forces across larger scales and higher Reynolds number regimes representative of real operating conditions, and (iii) validating numerical models against field-scale measurements to enhance their reliability and scalability.

1.5 Outline of the thesis

This thesis focuses on the analysis of the impacts of the geometric parameters and operational conditions on the turbine performances and structural loads of VATs. The parameters, including the turbine solidity (defined by the number of blades, chord length, and turbine radius), Reynolds number, blade-pitch angle, and tip-speed ratio (defined by stream speed and turbine rotational velocity), are considered. The numerical turbine configuration models will be built using a validated and reliable approach, and the accuracy will be assessed through comparison with field experimental data for both the wind and marine current applications.

The thesis is structured into three main areas, namely the introduction, numerical modelling, and findings, and is organized into eight chapters as outlined below.

- **Chapter 1 – Introduction**: provides the background, conversion technology, research gaps, and main aims of the thesis.
- **Chapter 2 – Turbine modelling techniques**: presents a basic theory of the dynamics of VATs and simulation turbulence.
- **Chapter 3 – Method**: describes a method for developing the numerical models. The method consists of grid generation, boundary conditions, and solver settings. In addition, the numerical models are verified by grid and time-step independency studies, and validated by the field experimental data, which ensures the modelling setup correctly and has the right modelling, respectively.

- **Chapter 4 – Results and discussion**: Presents a detailed analysis of the turbine performance and unsteady structural forces under varying design parameters and operating conditions, and discusses the observed fluid dynamics phenomena and the findings derived from the studies.
- **Chapter 5 – Conclusions**: Presents the main contributions of the thesis.
- **Chapter 6 – Summary of papers:** Present a brief summary of each paper in the thesis, and include the author's contribution to each paper.
- **Chapter 7 – Future work suggestions:** List of possible studies that can be performed by future PhD students within the topic.
- **Chapter 8 – Sammanfattning på svenska**: Summary written in Swedish.
- **Acknowledgements**: Expresses thanks to the supporters.
- **References**: Contains all references cited in the thesis.

2. Turbine modelling techniques

2.1 Fluid dynamics of vertical-axis turbines

To understand the basic aerodynamics of a vertical-axis turbine, a two-dimensional blade element is presented in Figure 2.1. A position of the blade is defined by the azimuthal angle (θ), where the upstream region is defined from θ = 0° to 180°, and the downstream one is from 180° to 360°.

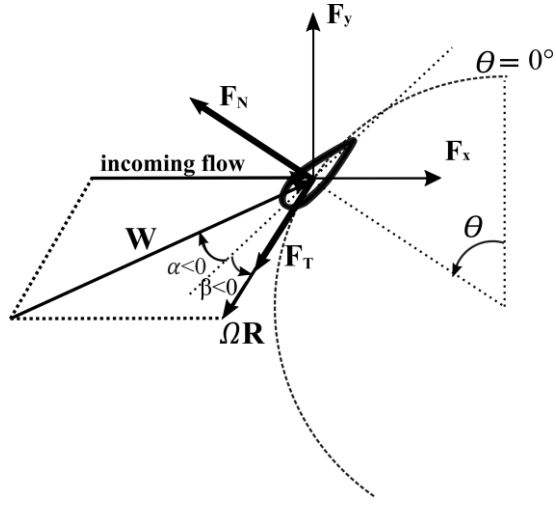


Figure 2.1. Direction of pitch angles, azimuthal angles, and forces (adapted from Paper II).

Two velocity components acting on the blade are the incoming flow velocity at the turbine blade ($\vec{V}_{in}$), and the rotational velocity ($\vec{V}_R$). The relative velocity ($\vec{W}$) is defined as below:

$$\vec{W} = \vec{V}_{in} - \vec{V}_R \quad (1)$$

Angle of attack (α) is an angle between the relative velocity and the chord line. The angle of attack depends on the azimuth angle and the pitch angle (β), and will therefore vary throughout a revolution. The angle of attack is the most important parameter for determining the lift and drag coefficients of a blade.

$$\alpha = -arctan\left(\frac{sin\theta}{TSR+cos\theta}\right) - \beta \quad (2)$$

Here, TSR is the tip speed ratio, i.e. the ratio between the blade speed and the flow speed, which is defined as

$$TSR = \frac{\Omega R}{V_\infty} \tag{3}$$

A second important parameter to characterize the lift and drag force created by the blade is the Reynolds number, which is defined as follows.

$$Re_c = \frac{\rho \Omega R c}{\mu} \tag{4}$$

Another important turbine parameter is the solidity, which basically is the ratio between the area covered by the blades and the total turbine area.

$$\sigma = \frac{\mathrm{Nc}}{2\mathrm{R}} \tag{5}$$

The two-dimensional (2D) lift and drag are estimated as

$$L_{2D} = \frac{1}{2}\rho\, c\, W^2\, C_L(\alpha, Re_c) \tag{6}$$

$$D_{2D} = \frac{1}{2}\rho\, c\, W^2\, C_D(\alpha, Re_c) \tag{7}$$

These forces could be used to calculate the forces in the normal and tangential directions by simply rotating the coordinate system with the flow direction. However, when using CFD, the forces are instead calculated by integrating the pressure and shear stress over the blade surface. Here, one will typically obtain the forces in the x and y directions instead. Hence, the instantaneous forces in the normal direction (F_{ni}) and in the tangential direction (F_{ti}) acting on a single blade (i) with the blade height (H) are calculated by Eq. 8:

$$F_i = (F_{ni}, F_{ti}) = H \begin{pmatrix} F_{xi} \\ F_{yi} \end{pmatrix} \begin{pmatrix} -sin\theta & +cos\theta \\ -cos\theta & -sin\theta \end{pmatrix} \tag{8}$$

On the turbine level, we are both interested in the forces on the turbine, and the torque on the turbine. The instantaneous thrust and lateral force acting on the whole turbine are calculated as

$$F_x = H \sum_{i=1}^{N} F_{xi} \tag{9}$$

$$F_y = H \sum_{i=1}^{N} F_{yi} \tag{10}$$

To calculate the average thrust coefficient (or the coefficient of the lateral force), we can average the total force for the turbine over one revolution according to the equation below.

$$C_{x,(y)} = \frac{\frac{1}{2\pi}\int_{\theta}^{\theta+2\pi} H \sum_{i=1}^{N} F_{x_i\,(y_i)}(\theta)\, d\theta}{0.5\rho H D V^2} \tag{11}$$

The total instantaneous torque acting on the whole turbine is:

$$T(\theta) = R \sum_{i=1}^{N} F_{ti} = RH \sum_{i=1}^{N} \left(-F_{xi} cos\theta - F_{yi} sin\theta\right) \tag{12}$$

This allows us to calculate the average torque in a similar way as before by calculating the average over one revolution.

$$T_{av} = \frac{1}{2\pi}\int_{\theta}^{\theta+2\pi} T(\theta)\, d\theta \tag{13}$$

This allows us to define the average torque coefficient as

$$C_T = \frac{T_{av}}{0.5\rho H D V^2 R} \tag{14}$$

We can also define the torque amplitude using the difference between the largest and smallest torque the turbine experiences during one revolution as

$$C_{|T|} = \frac{T_{max} - T_{min}}{0.5\rho HDV^2R} \tag{15}$$

Finally, the power coefficient of the turbine is calculated as follows.

$$C_P = \frac{\text{turbine power}}{\text{total power of flow}} = \frac{\mathrm{P_{turbine}}}{\frac{1}{2}\rho \mathrm{V^3HD}} = \frac{\frac{1}{2\pi}\int_{\theta}^{\theta+2\pi} \Omega \mathrm{T}(\theta)\mathrm{d}\theta}{0.5\rho \mathrm{V}^3 HD} = \frac{\Omega T_{av}}{0.5\rho \mathrm{V}^3 HD} \tag{16}$$

2.2 CFD modelling

To conduct the numerical analysis of the VATs, the turbines must be modelled using appropriate mathematical models, such as streamtube, vortex, actuator line models, or CFD [43,44]. In the streamtube models, the flow field is divided as a grid of independent streamtubes. The idea is to solve the flow velocity for each streamtube independently. This is achieved by first dividing the blades into segments in the vertical direction (one segment for each streamtube), and in the horizontal direction, each streamtube represents a given position along the revolution. The model will then calculate the blade forces using the flow velocity from external lift and drag coefficients, and the velocity can then be calculated from the force through momentum conservation. One typically uses one set of streamtubes for the upwind part, and then a second set of streamtubes for the downwind part [45]. The models assume flow in steady state, and the full description of the flow through the entire turbine is not obtained. In the vortex models, each blade is represented as a line or surface carrying bound vorticity, with wake evolution governed by vortex dynamics; viscous and boundary-layer effects are neglected, and instead, the forces are either calculated by assuming ideal airfoils (or hydrofoils), or calculated the same way as for the streamtube models, using external lift and drag coefficients [46,47]. In the actuator line models, each blade is modelled as a rotating line discretized into multiple force points along the span, where aerodynamic forces are distributed into the surrounding flow field [48]. All of these approaches require the external fluid dynamics input data, such as lift and drag coefficients as functions of angle of attack and Reynolds number. These data are typically obtained from simulation tools, or experiments such as in wind tunnels, XFOIL simulations, or two-dimensional computational fluid dynamics analyses [49–52]. A key limitation is that wind tunnel measurements are typically conducted on stationary airfoils, whereas vertical-axis turbine blades undergo a curved rotational motion that produces fundamentally different flow characteristics. This discrepancy becomes particularly significant under stall conditions, where dynamic stall exhibits strong azimuthal dependence. Unlike the streamtube, vortex, and actuator line models, which

require some external measurement data for the employed foils, the CFD simulations solve the full flow field and do not require external data, including the development of boundary layers [44]. This capability allows the CFD to provide a detailed and accurate prediction of stream characteristics around the turbines. With these advantages, the CFD is selected as the primary simulation tool for the turbine studies in this thesis.

CFD turbulence model

Three main approaches in turbulent CFD computations are Direct Numerical Simulation (DNS), Large Eddy Simulation (LES), and Reynolds-averaged Navier-Stokes simulations (RANS) [44,53]. DNS solves the full Navier-Stokes equation without employing any turbulent modelling, from the smallest to the largest eddies, and requires 3D simulations as the turbulence is very three-dimensional. This results in extremely high computational cost, and is beyond industrial applications. LES solves large-scale turbulent eddies directly (where the turbulence structures are larger than the mesh size), and models the smaller eddies [54]. Although it is less computationally demanding than DNS, it still requires significant computational resources, especially near walls where fine meshes are needed. In contrast, RANS resolves the time-averaged flow and models the entire spectrum of turbulence using a turbulence model without the need of resolving the turbulence eddies [44]. It allows for relatively coarse computational meshes and significantly lower computational cost than the rest of the approaches. Therefore, the RANS approach has been employed in the present work. The RANS equations use the time-average fields for both velocity and pressure.

The flow process of incompressible fluid motion obeys the Navier-Stokes equations, which are as follows.

$$\frac{\partial \mathrm{u_i}}{\partial \mathrm{x_i}} = 0 \tag{17}$$

$$\frac{\partial \mathrm{u_i}}{\partial \mathrm{t}} + \mathrm{u_j}\frac{\partial \mathrm{u_i}}{\partial \mathrm{x_j}} = -\frac{1}{\rho}\frac{\partial \mathrm{p}}{\partial \mathrm{x_i}} + \frac{\partial}{\partial \mathrm{x_j}}\left(\nu\frac{\partial \mathrm{u_i}}{\partial \mathrm{x_j}}\right) + \mathrm{S_i} \tag{18}$$

u_i (i and j = 1,2,3) is the velocity component; p represents the pressure; ν denotes the kinematic viscosity; ρ is the density of the fluid; and S_i denotes the source term.

The turbulence model employed is the two-equation k-ω SST model, which is a hybrid model combining elements of the k-ω and k-ε turbulence [55]. It performs well in various flow regimes, including both boundary layers and free shear layers. More details about the turbulence model assessment can be found in the work ref. [56]. The model is particularly beneficial in the case of adverse pressure gradients and separated flow. The Curvature Correction was implemented for the computation of eddy-viscosity models, which is introduced as a modification of the turbulence production term to account for the effects of the streamline curvature and the system rotation [57]. In order to

avoid the build-up of turbulent kinetic energy in the stagnation regions, the Production Limiter features are also enabled for the excessive generation of the turbulent energy [58].

3. Method

The CFD method consists of three major steps: preprocessing, solver execution, and postprocessing. First, the preprocessing is to create geometries and generate meshes. This step also consists of defining the flow properties and the boundary conditions. Second, the solver execution is to solve the governing equations iteratively to obtain predicted results. Third, the postprocessing is to analyze and visualize the predicted outcomes, which can gain insights into the fluid dynamics behaviors in different forms, such as velocity contour plots, pressure distributions, or streamlines. In addition, the accuracy of CFD is an important matter, so model verification and validation are required to assess the modelling errors.

3.1 Computational domain and grid generation

The computational domain is generated by means of ANSYS®/Design Modeler software. The computational domain is made of two zones (Figure 3.1). The first zone is a circular inner domain having a diameter (1.5D) and rotating around the turbine rotational axis with its angular velocity (Ω). The second one is a rectangular outer domain, which is stationary. The inlet flow has a uniform velocity and zero pressure gradient.

The rotating and the stationary zones are connected by a sliding mesh interface (*Figure 3.2*). The finer meshes by generating designated-sized meshes are employed in the numerical simulations. The finer meshes consist of the circular areas applied around every single blade, and one enclosing the rotating area. The circular areas have the centre point located at the origin of their foil centres (0.5c), and have the radius equal to the chord length. The enclosing area over the rotating zone is stretched towards the downstream region of the domain to better catch the velocity gradients in the turbine's wake.

The computational grid is created by the unstructured mesh for the global domains and influence bodies, and by the quadrilateral mesh for the boundary layer grid close to the walls.

For a turbine operating in an open-site environment, it is evident that blockage effects, commonly encountered in wind tunnels or water tanks, are naturally absent. As defined by Garrett and Cummins, the blockage ratio (ϵ) is

given by a ratio between the cross-sectional area of the turbine (A_s) and channel (A_c), $\epsilon = A_s/A_c$ [59]. When the channel width is reduced, the turbine performances will increase, and the turbine will shift its optimum at a higher TSR [34,60]. To eliminate any such influence caused by lateral boundaries, a sufficiently large computational domain is needed [61]. In the present work, the overall domain is characterized by an inlet length equal to d_i, an outlet length equal to d_o, and a width (in the orthogonal direction with respect to the incoming flow). Those domain parameters can be found in Table 3.1. Accordingly, no correction for the blockage is applied to either the field test data or numerical simulations.

Table 3.1. *Computational-domain settings.*

Parameter	**Wind VAT**	**Marine current VAT**
Inlet distance, d_i, [m]	54R	60R
Outlet distance, d_o, [m]	110R	120R
Lateral distance, [m]	120R	110R

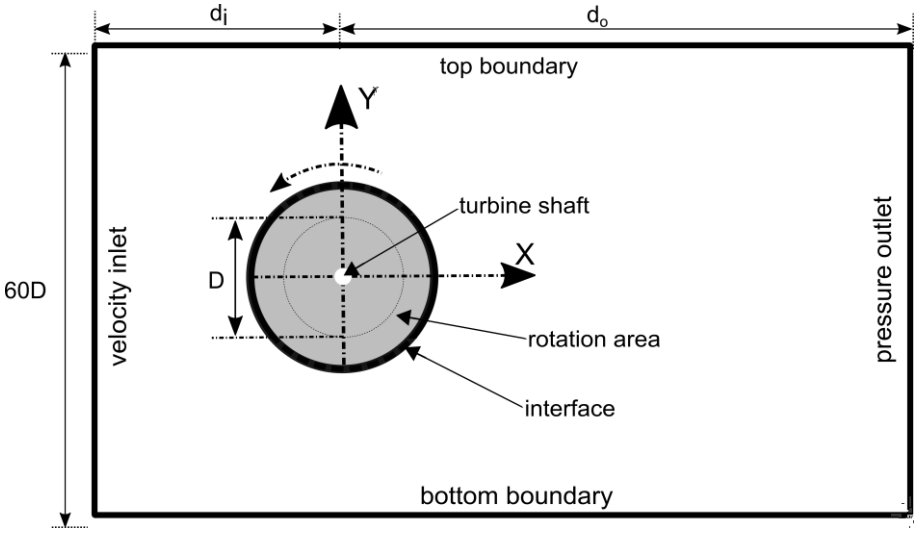


Figure 3.1. Computational domain. Adapted from Paper IV.

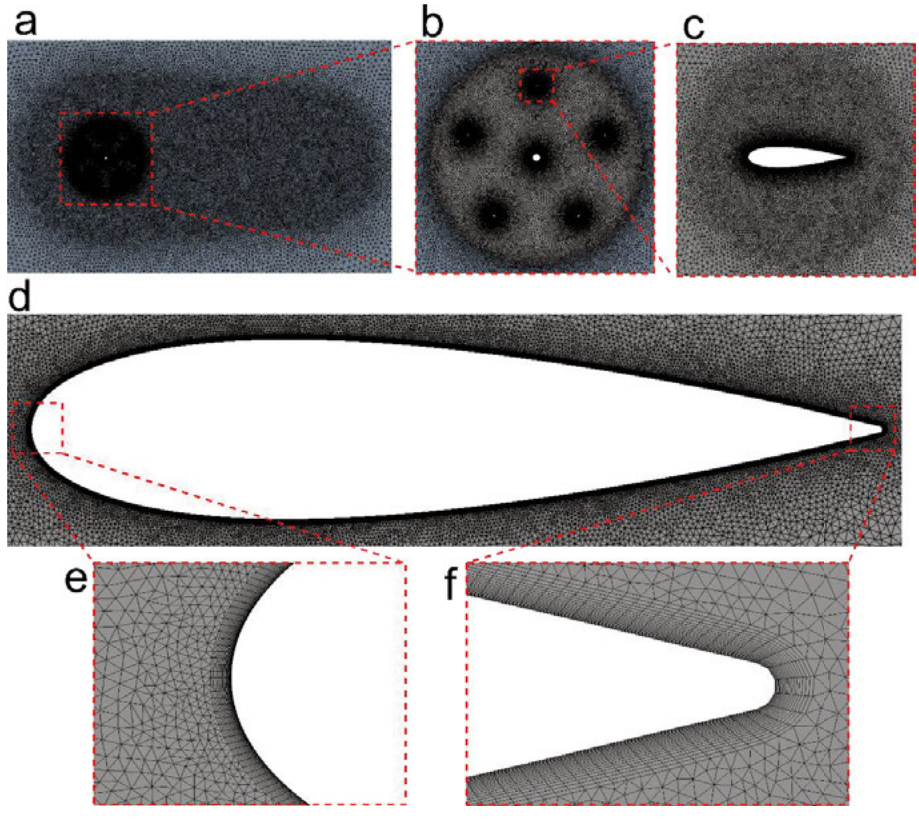


Figure 3.2. Computational grid: a) turbine with influence body; b) turbine; c) blade-foil with influence body; d-f) computational grid near blade-foil (d), leading edge (e), trailing edge (f). Adapted from Paper III.

3.2 Solver settings

As discussed in Chapter 2, the unsteady flow field around the VATs is numerically simulated using the unsteady Reynolds Averaged Navier-Stokes (U-RANS) equations. These equations involve the time-averaging of continuity and momentum equations for the flow. The U-RANS equations are solved with a coupled algorithm, employing a second-order upwind scheme for momentum and turbulence, along with a bounded second-order implicit scheme for the transient formulation.

The turbulence modelling is modelled using the SST k-omega turbulence model. The boundary conditions at the inlet and outlet are uniform velocity and Gauge pressure, respectively. The inlet velocity is kept constant for all numerical simulations, while the outlet is defined with pressure equal to 0 Pa. The top and bottom boundaries are free-slip walls that have the shear stress equal to zero since they are far from the turbine. On the blade surfaces, the flow moves with zero relative speed, therefore, the surfaces have no-slip walls.

3.3 Model verification and validation

Verification and validation for the numerical models are essential steps of the modelling process, ensuring both the correctness of the computational implementation and the reliability of the predicted results. The verification aims to confirm that the settings of the model are correct through the independence studies on the grid and time step (reported as azimuthal angle variation). The validation, on the other hand, focuses on assessing whether the numerical model accurately represents the physical system. This is achieved through comparison of simulation results with experimental data. In this work, validation is primarily performed using field measurement data. Comprehensive details of the verification and validation are provided in Chapter 3 of Paper II, in Chapters 2 and 3 of Paper III, and in Chapter 3 of Paper IV.

A brief overview of the turbines employed for the verification and validation is presented here. Two representative configurations are considered: a three-blade wind turbine and a five-blade marine current turbine. The key parameters of these reference configurations are found in **Table 3.2**. For the wind turbine case, the validation focuses on the prediction of unsteady aerodynamic loads, especially the normal force acting on the blade. In contrast, for the marine current turbine, the emphasis is on evaluating overall turbine performance, typically through the power coefficient.

Table 3.2. *Parameters of reference turbines.*

Parameter	Wind VAT	Marine current VAT
Number of blades, [-]	3	5
Blade profile, [-]	NACA 0021	NACA 0021
Inlet speed, [m/s]	6.4	1.4
Blade height, H [m]	5	3.5
Turbine radius, R [m]	3.24	2.94
Chord length, c [m]	0.25	0.18
Shaft radius, [m]	0.1	0.15

First, grid independency with respect to the refinement levels is evaluated. Three kinds of meshes for grids are employed: a coarse grid, a medium grid, and a fine grid. The number of cells of each case is found in **Table 3.3**. The outcomes from the three different meshes are presented in *Figure 3.3*a and *Figure 3.4*b. It looks clear that the difference in outcome when comparing the fine grid from the medium meshes is much smaller than that from the coarse meshes. For example, in *Figure 3.4*a, the blade-average torque at TSR of 3.1 is about 0.35% and 4.23% difference in the medium and coarse meshes compared to the fine meshes, respectively. Therefore, the medium grids are chosen for all the CFD simulations in the present work.

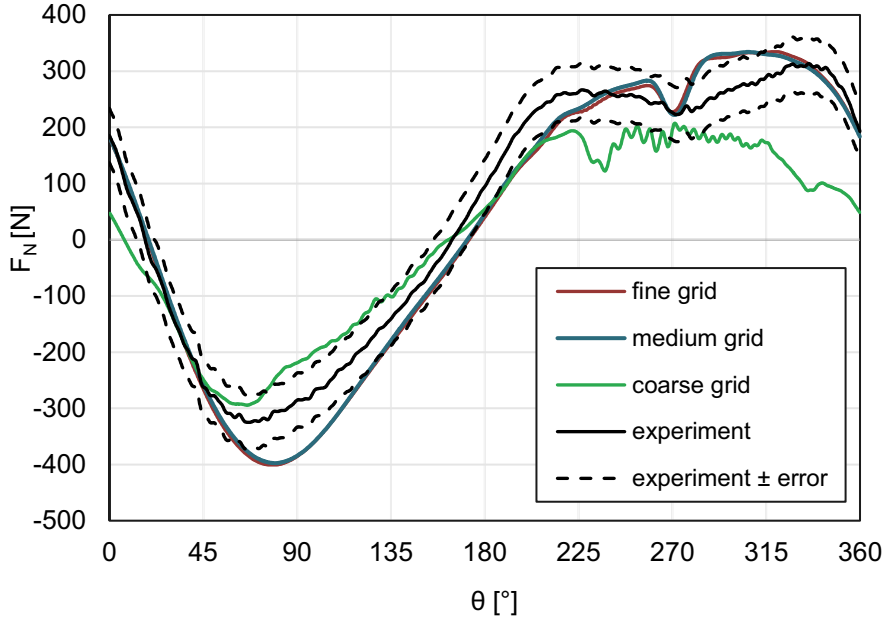
Table 3.3. *Number of cells in the different grids for the wind and marine current VAT.*

Mesh	Wind VAT	Marine current VAT
Fine	850 000	1110 000
Medium	424 000	500 000
Coarse	200 000	250 000

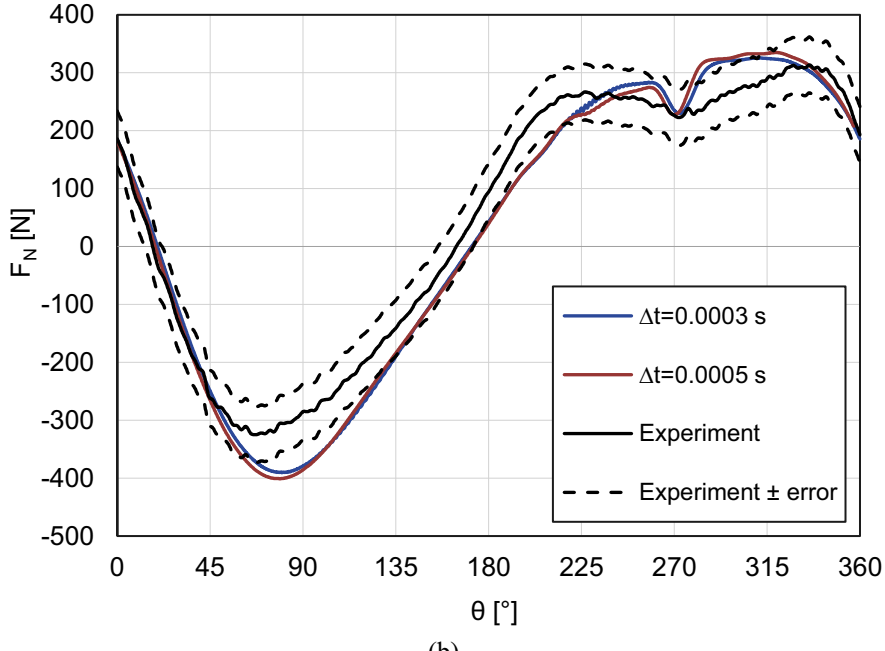
Second, the impact of time or azimuthal steps was examined with different values. The time-step verifying process is done similarly to the mesh independency study. The results are presented in *Figure 3.3*b and *Figure 3.4*b. It can be seen that the results are close to each other between $\Delta\theta = 0.1°$ s and $\Delta\theta = 0.2°$ (that corresponds to $\Delta t = 0.0003$ s and $\Delta t = 0.0005$ s, respectively) in Figure 3.3b, and between $\Delta\theta = 0.1°$ and $\Delta\theta = 0.15°$ in Figure 3.4b. Hence, $\Delta\theta = 0.2°$ and $\Delta\theta = 0.1°$ are chosen for the numerical models of the 3-blade and 5-blade turbine, respectively.

Finally, the numerical model is validated with the field experimental data. To have more information about the experiment setup and the data collected, please read more in references [29,31,33,62–64]. The validation results are shown in *Figure 3.3*b and *Figure 3.4*c. In *Figure 3.3*b, two data sets of field wind experimental data are shown: one represents measurements without errors, namely experiment (black-solid lines), and the other includes measurement uncertainty with errors, labelled as experiment ± error (black-dash lines).

In *Figure 3.4*b, the empirical correlation derived from the experimental resistant torque is applied to the hydrodynamic CFD torque output. The relation factor was included to account for the influence of such turbine struts [65]. Regarding the normal force shown in *Figure 3.3*b, the numerical prediction is mostly within the limited range of the experiment. Figure 3.4c shows a suitable agreement in the power coefficient between the simulation and measurement, especially at TSR = 3.0 – 3.5 near the optimal TSRs, as discussed in Chapter 4, where the discrepancy is minimal. In general, the numerical simulations demonstrate reliable predictive capability for the VATs' outcomes.

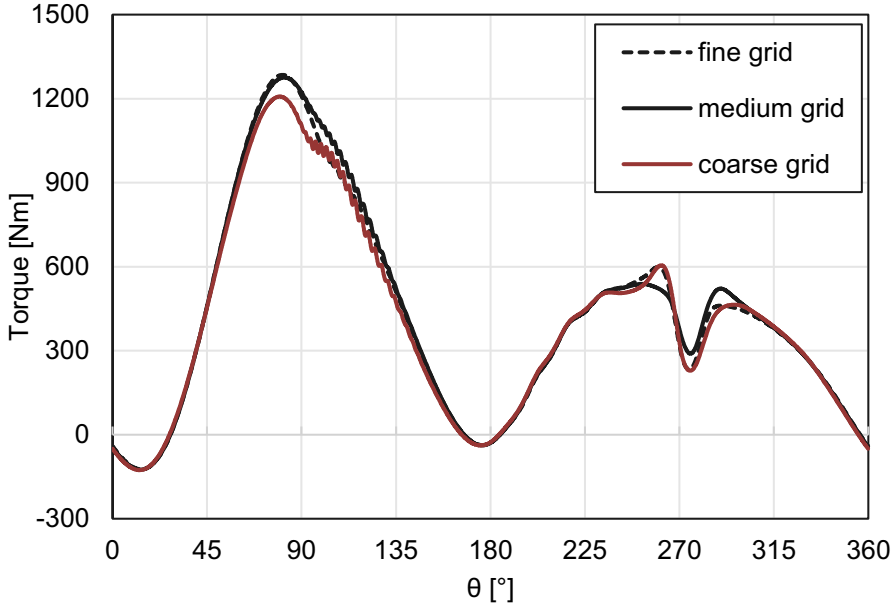


(a)

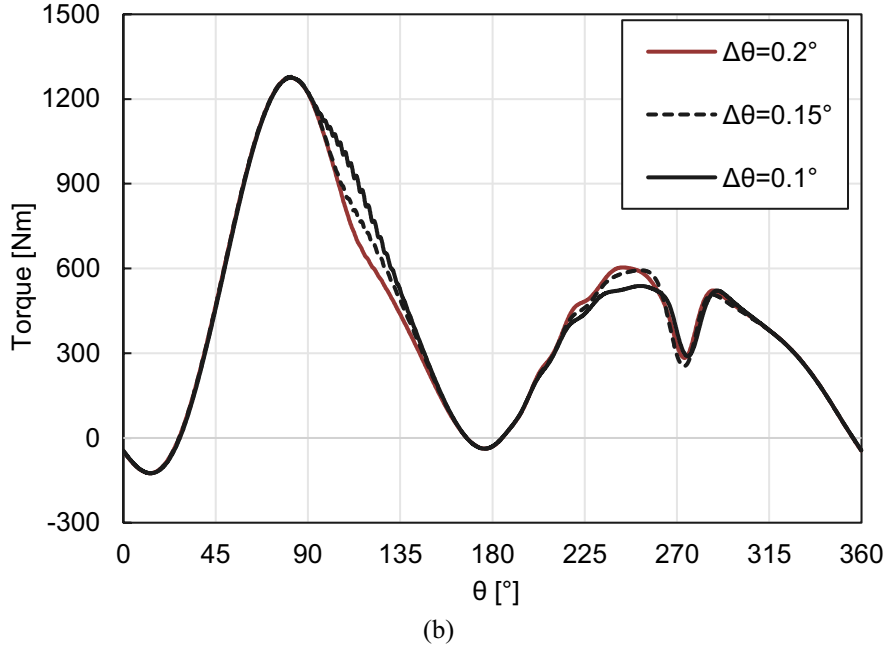


(b)

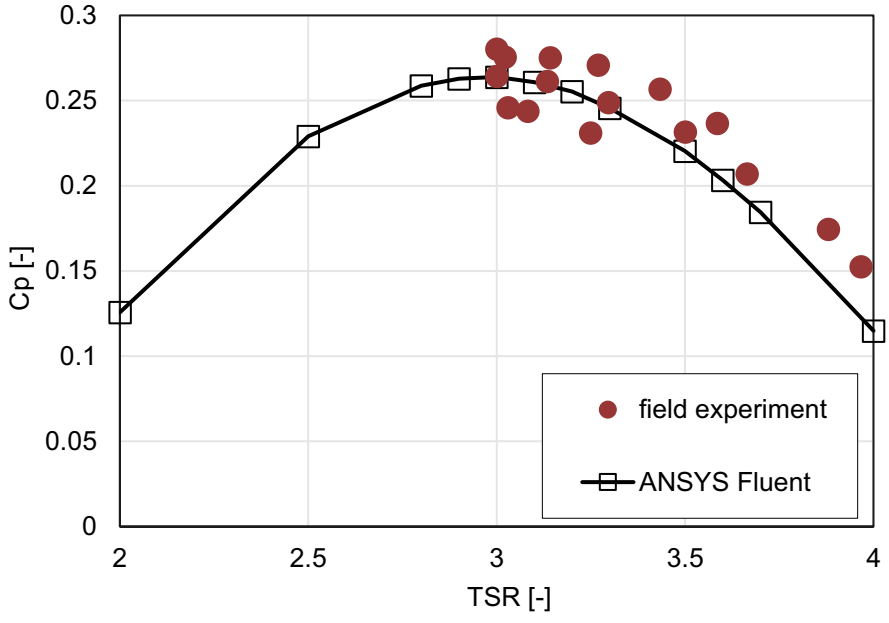
Figure 3.3. Grid independency (a), azimuthal-step independency and validation (b) for the three-blade wind VAT at U_0 = 6.4 m/s. Adapted from Paper II.


1500
1200
900
600
300
0
-300
Torque [Nm]
0
45
90
135
180
225
270
315
360
θ [°]
fine grid
medium grid
coarse grid

(a)


1500
1200
900
600
300
0
-300
Torque [Nm]
0
45
90
135
180
225
270
315
360
θ [°]
Δθ=0.2°
Δθ=0.15°
Δθ=0.1°

(b)



(c)

Figure 3.4. Grid and azimuthal-step independency (a-b), and validation (c) for the five-blade marine current VAT at U_0 = 1.4 m/s. Adapted from Paper III.

4. Results and discussion

This chapter presents a comprehensive analysis of turbine behavior under varying conditions. Section 4.1 focuses on the evaluation of unsteady aerodynamic forces, Section 4.2 examines the influence of blade pitch angles, and Section 4.3 investigates the effects of multiple design and operating parameters on turbine performance.

First, for the three-bladed wind turbine configuration, the numerically predicted structural loads are compared with field experimental data obtained from an open-air test site. A detailed analysis of the force characteristics is provided in Paper II.

Second, for the five-bladed marine current turbine, the effects of blade pitch angle on turbine performance are systematically investigated through an eleven-case parametric study, covering a wide pitch range from −5° to +5°, as presented in Paper III.

Finally, a parametric analysis is conducted using sixteen turbine configurations to evaluate the combined influence of geometric parameters—such as blade number, rotor radius, and chord length—and operational conditions, including flow velocity, tip-speed ratio, and Reynolds number. The corresponding impacts on both turbine performance and structural loading are discussed in detail in Papers IV and V.

4.1 Structural loads

The primary objective of this section is to compare the predicted structural loads of the wind VATs with corresponding experimental measurements in order to assess the accuracy of the numerical model. Table 4.1 summarizes the key parameters of all case studies, including geometric properties, flow conditions, and operational settings such as the tip-speed ratio (TSR). The comparative analysis will ensure accuracy between the numerical simulations and the experimental data, thereby providing a reliable insight into phenomena occurring during the revolution.

Three representative tip-speed ratios (TSRs = 1.84, 3.74, and 4.57) are considered. These values are selected to cover a range of operating conditions, from low to relatively high TSR, allowing for a thorough evaluation of model operation under different flow regimes.

Table 4.1. *Case-study parameters of the wind VAT.*

Parameter	Wind VAT
Wind speed, [m/s]	4.9 – 7.4
Turbine radius, R [m]	3.24
Number of blades, N [-]	3
Blade profile, [-]	NACA 0021
Blade length, H [m]	5
Chord length, c [m]	0.25
Blade pitch angle, [°]	+2
Rotor solidity factor, Nc/2R [-]	0.115
Shaft radius, [m]	0.1
Tip-speed ratio, TSR [-]	1.84 – 4.57
Chord-based Reynold number, Re_c [-]	$2.2\times10^5 – 3.7\times10^5$

Normal forces

Figure 4.1 shows the variation of the normal force against the azimuthal angle over one complete revolution at TSR = 1.84, and V = 6.4 m/s. Fluctuations do appear in the numerical simulation results and, similarly, though less pronounced, the variations are also observed in the experimental data at the same TSR. These discrepancies could be explained by the limitations of the two-dimensional modelling approach, which tends to have reduced accuracy in capturing the flow separation, especially at the low TSR, where the blade has stalled, and stall-induced vibrations are evident. Moreover, the 2D simulations assume a uniform force distribution along the blade heights, which does not fully represent the actual three-dimensional flow effects. Despite these limitations, the overall agreement between the simulations and experiments is reasonably matching. At this TSR, the numerical model slightly overestimates the peak normal force in the upwind region. According to the experimental data (solid black line) in Figure 4.1, the maximum normal force occurs at an azimuthal angle of approximately 73°, while the CFD predicted the peak at the angle of 58°, resulting in a 24% difference between the numerical and experimental values.

Figure 4.2 presents the outcomes obtained at TSR = 3.74 and V = 5.9 m/s. At this operating condition, the fluctuations observed in the normal force at lower TSR (TSR = 1.84) are significantly reduced, resulting in a smoother force distribution. In addition, the drastic reduction in the normal force associated with the central tower becomes more clearly identifiable. Although the region of force reduction predicted by the simulation is narrower than that observed experimentally, it effectively indicates the azimuthal range where strong blade–structure interaction occurs, approximately between 247° and 292°. Overall, the CFD results demonstrate very good agreement with the experimental data in terms of the trend of the normal force produced correctly.

Finally, Figure 4.3 illustrates the results at the highest TSR, TSR = 4.57, and V = 4.9 m/s. In the upwind region, the CFD-predicted normal force remains within the experimental measurement bounds. Notably, the peak value of normal force occurs at approximately the same azimuthal position ($\theta \approx 90°$) in both cases. In the downwind region, the simulation predicts a sharp reduction in the normal force beginning at an azimuthal angle of approximately 260°, and extending to about 280°. In contrast, the experimental data indicate that the force reduction also initiates at 260°, but persists over a broader range, extending to approximately 320°.

In conclusion, the CFD simulations demonstrate good agreement with the experimental results across the TSRs examined. The numerical model successfully captures the key aerodynamic phenomena observed in the measurements. In particular, the influence of the turbine shaft, manifested as a pronounced reduction in normal force, is clearly reproduced, especially at moderate TSR conditions.

Among the operating cases considered, the turbine at TSR = 1.84 exhibits the highest level of force fluctuations, indicating increased unsteadiness at low rotational speeds. This behavior can be attributed to the inverse relationship between TSR and angle of attack, whereby lower TSRs correspond to higher angle of attack for a given free-flow velocity. At the elevated angle of attack, the flow is less likely to remain attached to the blade surface, leading to flow separation, stall, vortex shedding, and other unsteady aerodynamic effects. The phenomena related to dynamic stalls are very hard to predict numerically.

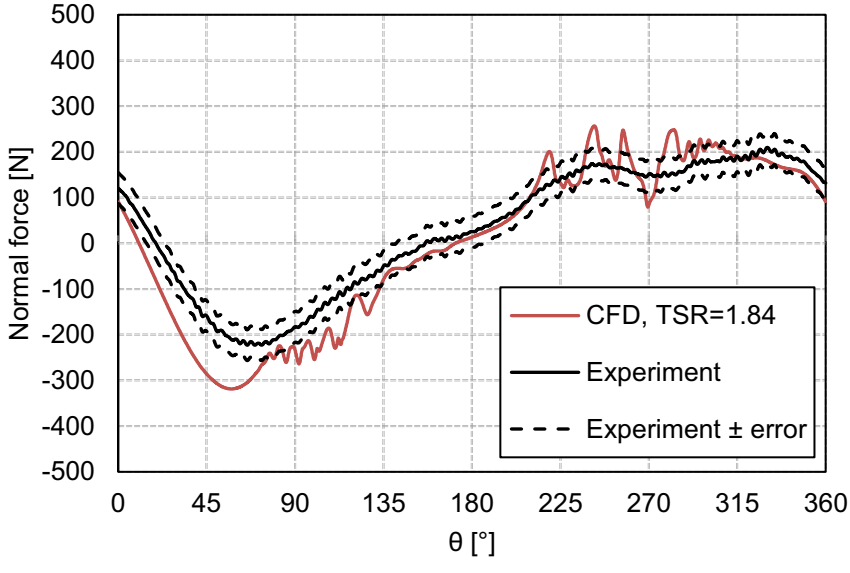


Figure 4.1. Normal force at TSR = 1.84. Adapted from Paper II.

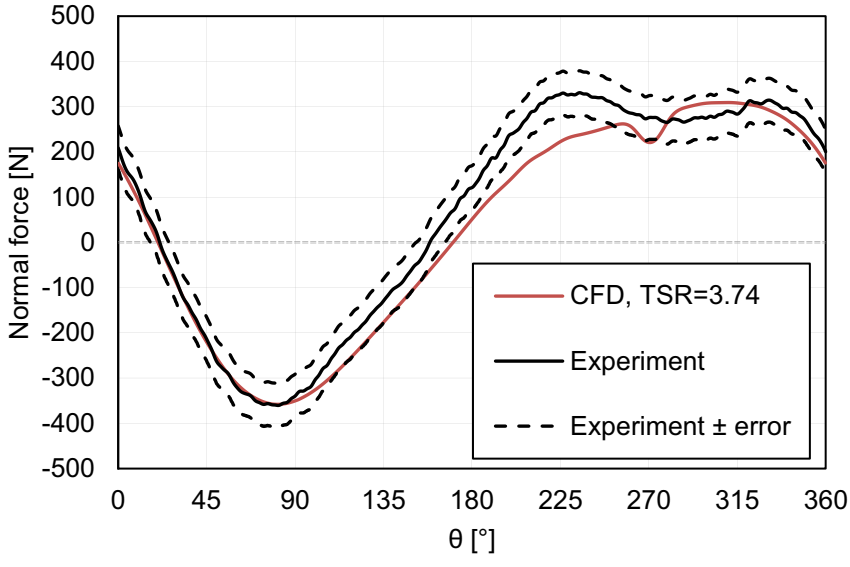


Figure 4.2. Normal force at TSR = 3.74. Adapted from Paper II.

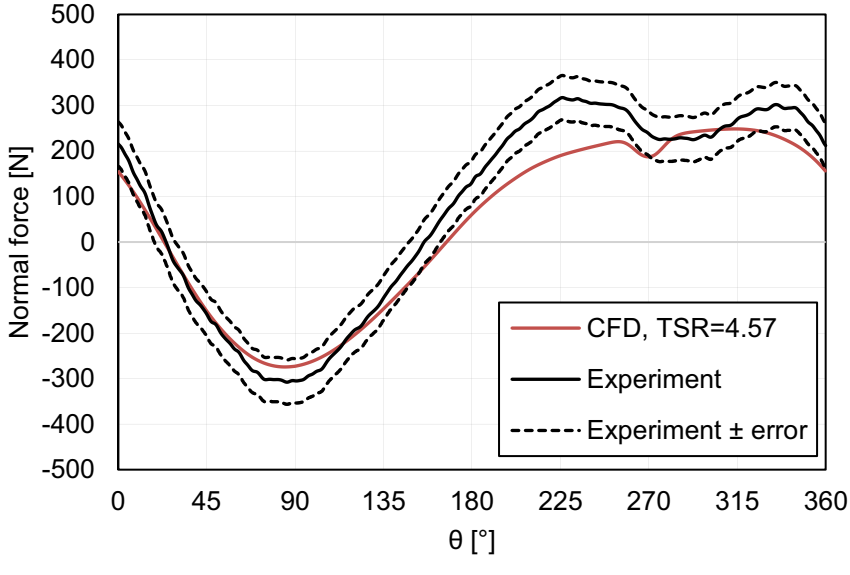


Figure 4.3. Normal force at TSR = 4.57. Adapted from Paper II.

Flow field around the turbines

To understand aerodynamic performance, the flow field around the turbine is examined. Figure 4.4 presents vorticity distributions at selected TSRs corresponding to positions where the blade experiences a peak value of normal force, torque, or angle of attack.

A key observation is that, across all examined TSRs, both the peak normal force and peak torque occur when the flow remains attached to the blade surface. In addition, a clear azimuthal offset is observed between these two maxima: the peak normal force occurs earlier in the rotational cycle, followed by the peak torque. This phase difference indicates that optimal aerodynamic performances are governed not only by the magnitude of the structural forces but also by their aerodynamic interaction.

The data in Table 4.2 further reveal a consistent trend with decreasing TSR. As the TSR decreases, the azimuthal positions corresponding to the peak normal force and peak torque shift progressively toward earlier angles in the rotation, compared to a position when the angle of attack is reaching its peak. At the lower TSRs, the earlier occurrence of peak loads suggests an increased likelihood of unsteady aerodynamic phenomena, such as dynamic stall during the post-peak behavior of blade motion. Those results provide further support for the flow instability identified at a low TSR in Figure 4.1.

Overall, these trends highlight the importance of appropriate TSR selection in turbine operation. Maintaining the turbine within an optimal TSR range can help ensure that peak loads occur under controlled, attached-flow conditions, or non-stalls, thereby improving efficiency, reducing structural loading fluctuations, and potential risks.

Table 4.2. *Maximum values of normal force, torque, and angle of attack at corresponding TSRs and V = 6.4 m/s.*

Maximum magnitude of	**TSR = 3.44**	**TSR = 3.0**	**TSR = 2.5**	**TSR = 2.0**
Normal force	$\theta = 79°$, $\alpha = -13°$	$\theta = 75°$, $\alpha = -14°$	$\theta = 67°$, $\alpha = -16°$	$\theta = 60°$, $\alpha = -17°$
Blade torque	$\theta = 90°$, $\alpha = -14°$	$\theta = 85°$, $\alpha = -16°$	$\theta = 74°$, $\alpha = -17°$	$\theta = 65°$, $\alpha = -18°$
Angle of attack	$\theta = 107°$, $\alpha = -15°$	$\theta = 110°$, $\alpha = -17°$	$\theta = 114°$, $\alpha = -22°$	$\theta = 120°$, $\alpha = -28°$

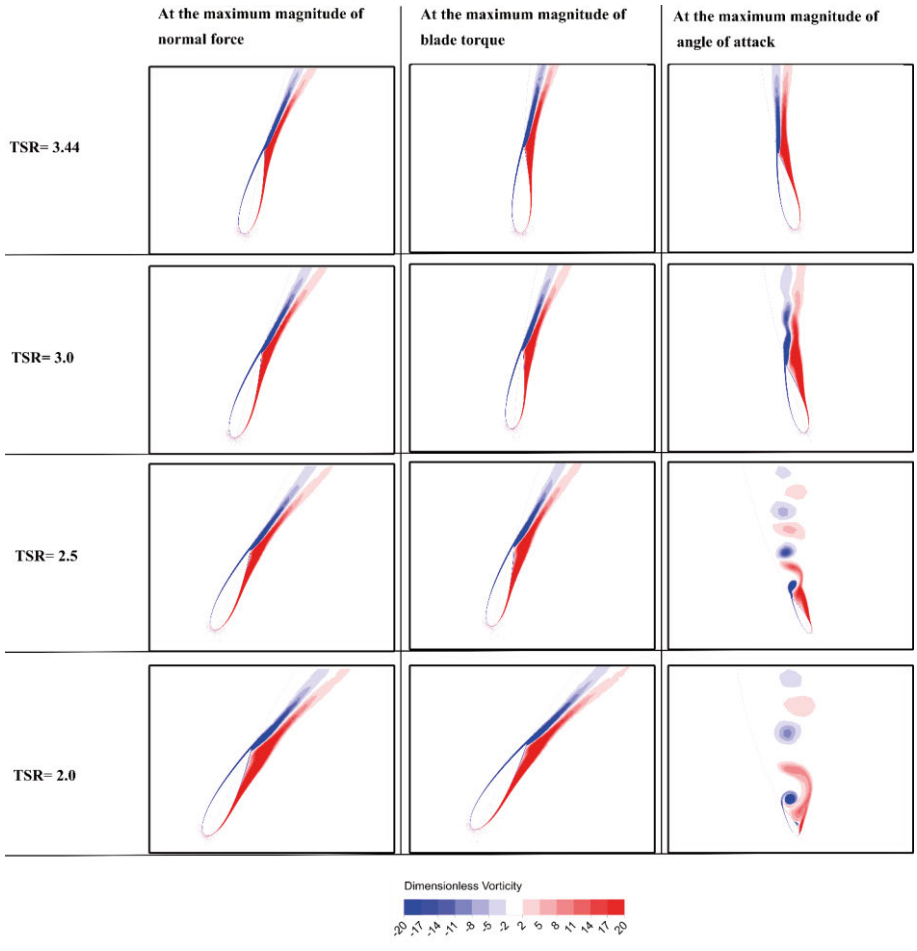


Figure 4.4. Vorticity field at different TSRs and U_0 = 6.4 m/s. Adapted from Paper II.

4.2 Blade-pitch angles

This section examines the influence of blade pitch angle on the hydrodynamic performance of vertical-axis marine current turbines (VAMTs) to identify the optimal pitch configuration. The analysis focuses on a five-bladed turbine operating under fixed pitch conditions, with blade pitch angles ranging from –5° to +5°. The definition of blade pitch and its orientation relative to the flow are illustrated in Figure 4.5. The key parameters of the marine current turbine system considered in this study are summarized in Table 4.3.

A zero-pitch turbine configuration is adopted as a reference case, as it has been previously validated against experimental measurements (see Section 3.3). This baseline provides a reliable benchmark for assessing the relative performance variations induced by different pitch angles.

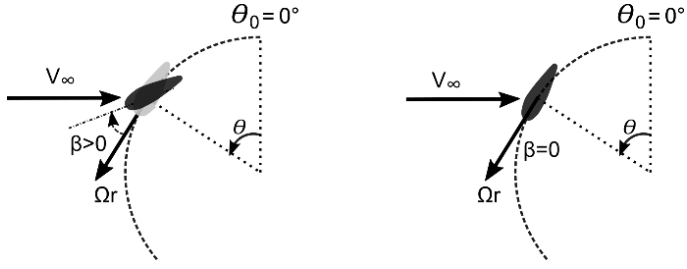


Figure 4.5. Direction of pitch angles.

Table 4.3. *Case-study parameters of marine current VAT.*

Parameter	**Marine current VAT**
Water speed, [m/s]	1.4
Turbine radius, R [m]	2.94
Number of blades, N [-]	5
Blade profile, [-]	NACA0021
Blade length, H [m]	3.5
Chord length, c [m]	0.18
Blade pitch angle, β [°]	± 5, ± 4, ± 3, ± 2, ± 1, 0
Rotor solidity factor, Nc/2R [-]	0.15
Shaft radius, [m]	0.15
Tip-speed ratio, TSR [-]	2.0 – 4.0
Chord-based Reynolds number, Re_c [-]	$5.0 \times 10^5 – 10.0 \times 10^5$

Power coefficients and instantaneous torques

Figure 4.6 illustrates the variation of the turbine power coefficient with blade pitch angle at optimal tip-speed ratios (TSR_{opt}). The result indicates that the power coefficient increases as the pitch angle varies from −5° to +2°, reaching a maximum at +2°, and subsequently decreases as the pitch angle increases

further to +5°. Overall, the turbines operating with positive pitch angles have higher efficiency compared to those with negative pitch angles. In addition, the slope of the power curve is steeper on the negative side of the pitch range, indicating a more pronounced sensitivity of performance to pitch variations in this region.

Figure 4.7 presents the instantaneous torque of a single blade as a function of azimuthal angle for a range of blade pitch angles. For negative pitch angle configurations, the maximum torque occurs earlier in the rotational cycle, prior to $\theta = 90°$, indicating premature flow separation or the onset of stall. Specifically, the peak torque is observed at $\theta = 78°$, 74°, 68°, and 63° for $\beta = -1°, -2°, -3°, -4°$, and $-5°$, respectively. In contrast, for positive pitch angles, the maximum torque generally occurs near $\theta = 90°$, reflecting more stable flow attachment and improved fluid dynamic performance.

Across all configurations, a noticeable reduction in torque is observed in the downstream region between $\theta = 255°$ and $\theta = 290°$, primarily due to the influence of the turbine shaft and wake interactions. This reduction is most pronounced at $\beta = -5°$, where the downstream torque deficit is greatest. Additionally, the upstream torque peak for this case occurs at the earliest azimuthal position among all configurations. Consequently, the $\beta = -5°$ case results in the largest power loss and the lowest power coefficient among the configurations examined.

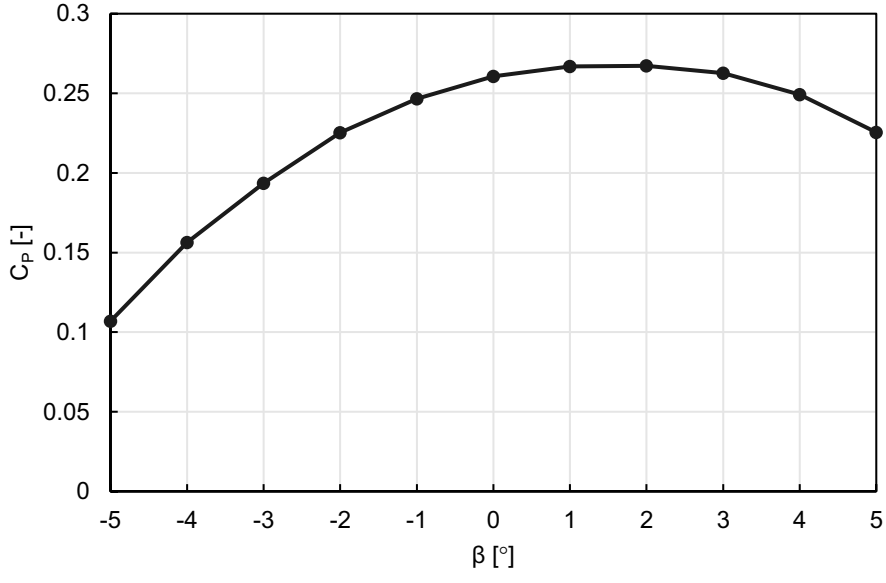


Figure 4.6. Optimal power coefficient against pitch angle. Adapted from Paper III.

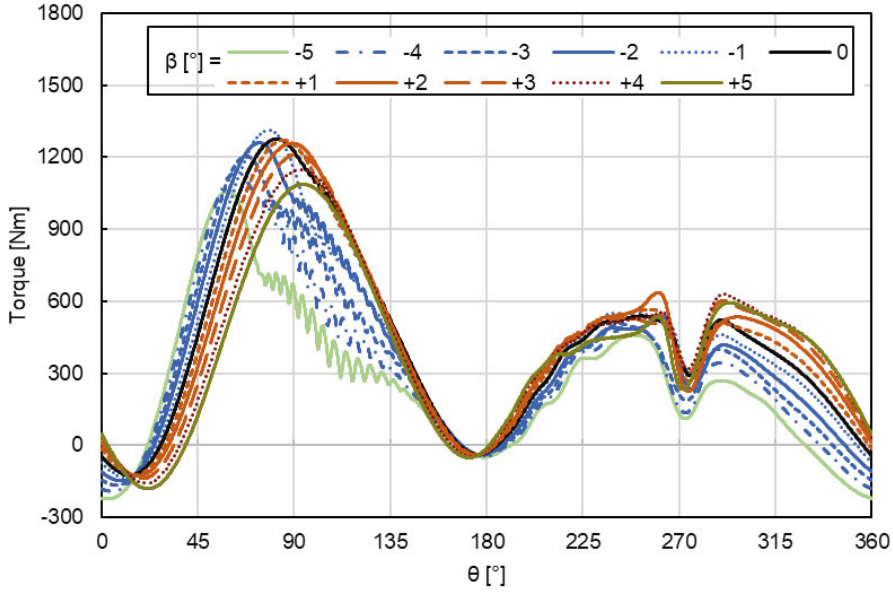


Figure 4.7. CFD instantaneous torque of a single blade against azimuthal angle at different pitch angles. Adapted from Paper III.

The highest power coefficient is identified at β = +2°. The hydrodynamic performance of this optimal configuration is further examined in Figure 4.8, which represents the instantaneous torques of a single blade against the azimuthal angles with a wide range of TSR (TSR = 2.0 – 4.0). For the low TSRs (2.0 and 2.5), fluctuations in torque occur in both the upstream zone (θ ≈ 90°–150°) and more significantly in the downstream zone (θ ≈ 200° – 280°), indicating unstable hydrodynamic behaviour. At the high TSR (TSR = 4.0), the upstream torque peak remains near θ ≈ 90°, and the flow is stable. The downstream, the position of the torque drop peak significantly shifts and extends over a wider range of θ ≈ 273° – 300° than the medium TSRs (3.0 – 3.5). For medium TSRs (3.0 – 3.5), the torque behaviour is more favourable. The upstream torque consistently reaches its maximum near θ ≈ 90°, while the downstream torque drop occurs around θ ≈ 270°. This pattern ensures relatively uniform torque generation during both the upstream and downstream halves of the rotation, minimizing unsteady loading and flow separation. As a result, moderate TSRs provide the highest average torque and most stable operation.

Figure 4.9 presents the turbine power coefficients (C_P) against TSRs (2.0 – 4.0) for five different pitch angles, β = –5°, –2°, 0°, +2°, +5°. The turbine with the optimal pitch angle of β = +2° has the maximum coefficient equal to 27%, which is 2.3% higher than that of the reference configuration with β= 0°. This indicates that a slight positive pitch enhances the turbine performance.

Furthermore, the optimal TSR for the maximum power slightly varies with the blade pitch. For negative pitches, the optimal TSR is 2.9 for both β = –5° and –2°, whereas for zero and moderate positive pitches (β = 0° and +2°) TSR_{opt} = 3.0. For the highest positive pitch examined (β = +5°), TSR_{opt} increases slightly to 3.1. This trend indicates that positive pitch angles allow the turbine to operate efficiently at slightly higher rotational speeds, likely due to improved flow attachment and delayed stall onset on the blades.

To further explain why positive pitch angles outperform negative ones, the flow field around the turbine is visualized for three representative pitch angles, β = +2°, 0°, and –2°, as shown in Figure 4.10. The flow contours for β = +2° exhibit fewer and less intense fluctuations compared to the other configurations, indicating more stable flow attachment over the blade surfaces. In this configuration, the incoming blades interact with weaker vortex structures shed by preceding blades, which reduces the likelihood of flow-induced instabilities and minimizes unsteady aerodynamic loads. In contrast, the zero and negative pitch cases display stronger vortex interactions and more pronounced fluctuations in the flow field, which can promote early boundary-layer separation, dynamic stall, and reduced torque generation.

These observations provide concrete evidence that moderate positive pitch angles enhance hydrodynamic stability by improving flow coherence and reducing unsteady loading. This understanding is critical for the turbine design process, as it allows designers to select pitch angles that optimize both power extraction and operational reliability.

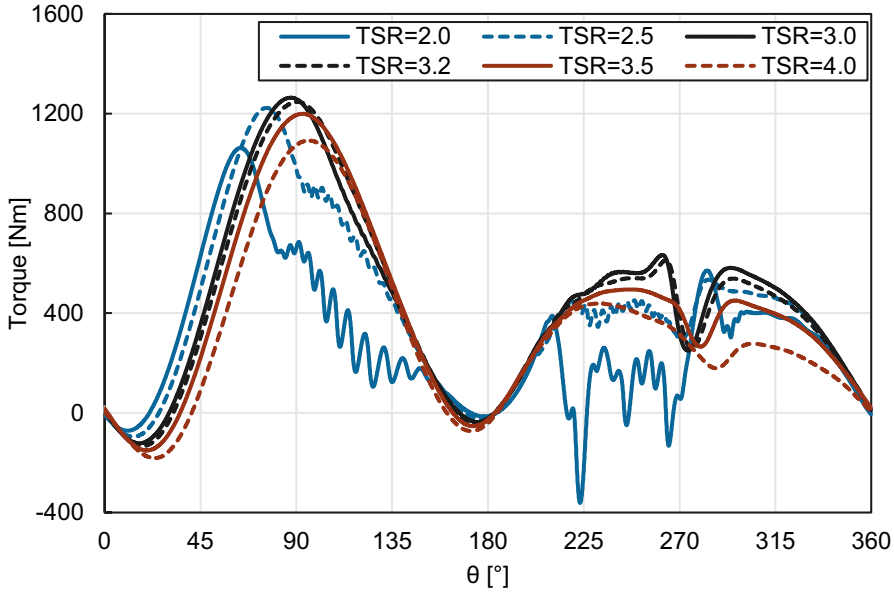


Figure 4.8. CFD instantaneous torque at the optimal β = +2. Adapted from Paper III.

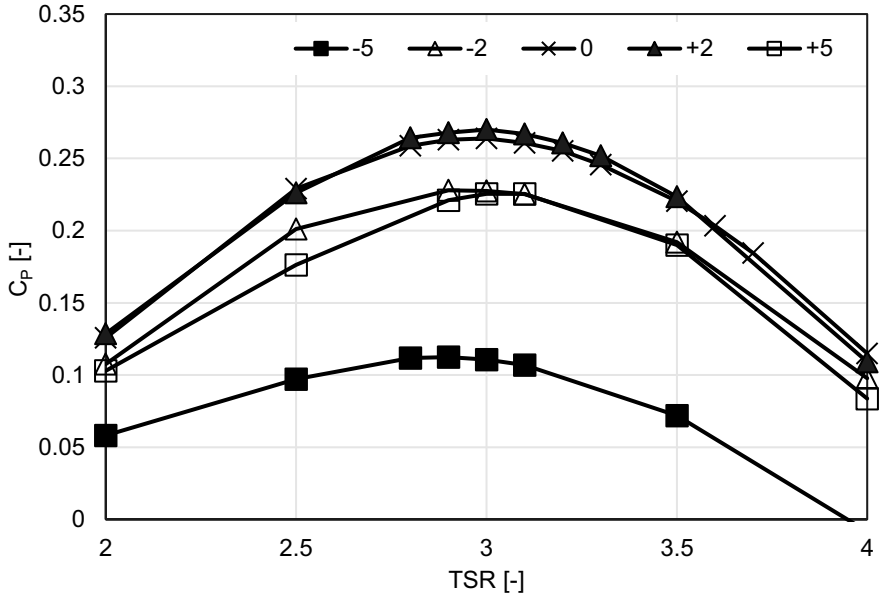


Figure 4.9. CFD power coefficient against TSR. Adapted from Paper III.

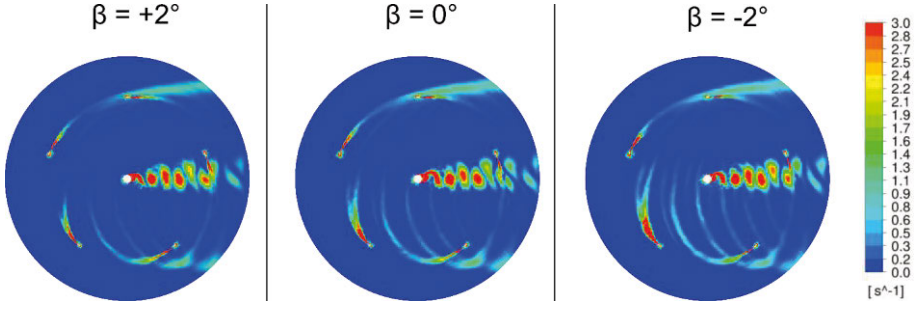


Figure 4.10. CFD vorticity of the turbine at different blade pitches. Adapted from Paper III.

4.3 Geometric parameters and operating conditions

Turbine solidity is a critical geometric parameter that significantly influences turbine hydrodynamics and overall performances of turbines. Changing in the solidity not only modify the local fluid dynamic characteristics around blades but also changes the entire turbine structure configuration and its operational behaviour. Therefore, the turbines with the different solidities should be considered entirely distinct designs rather than a simple variation of the single parameter.

To evaluate impacts of other parameters, such as number of blades, chord length, turbine radius on the turbine behaviour, it is necessary to maintain solidity as a constant. This ensures that observed phenomena during the turbine operation can be attributed directly from the parameters under the studies. In addition, the operating conditions, including inflow speed, rotational turbine speed, and Reynolds number, also strongly influence the outputs from the turbine systems. Therefore, a comprehensive design and optimization process must account for the combined effects of both geometric and operational parameters.

The present investigation focuses on numerical comparisons of marine current turbine performances among configurations in which solidity is held constant while other parameters are systematically varied. Three constant solidity values, $\sigma = 0.1$, 0.15, and 0.2 are considered. For each solidity, variations in the number of blades (N), blade chord length (c), turbine radius (R), tip-speed ratio (TSR), and chord-based Reynolds number (Re_c) are examined. In total, 16 turbine configurations are simulated, with their corresponding parameters summarized in Table 4.4. This approach provides a controlled framework for assessing the sensitivity of turbine performance to key design and operational variables.

Table 4.3. *Case-study parameters.*

Number of blades, N [-]	**Solidity, σ [-]**	**c/R, [-]**	**Chord, c [m]**	**Radius, R [m]**	**Re_c ($\times 10^5$)**	**TSR, [-]**
3-blade	0.102	0.068	0.20	2.94	5.5 − 12.5	2.0 − 4.5
	0.153	0.102	0.30	2.94	8.3 − 18.8	2.0 − 4.5
	0.204	0.136	0.40	2.94	11.1 − 25	2.0 − 4.5
	0.204	0.136	0.80	5.88	22.2 − 50	2.0 − 4.5
4-blade	0.102	0.051	0.15	2.94	4.2 − 9.4	2.0 − 4.5
	0.153	0.0765	0.225	2.94	6.3 − 14.1	2.0 − 4.5
	0.204	0.102	0.30	2.94	8.4 − 18.8	2.0 − 4.5
	0.204	0.102	0.60	5.88	16.8 − 37.6	2.0 − 4.5
5-blade	0.102	0.0408	0.12	2.94	3.3 − 7.5	2.0 − 4.5
	0.153	0.0612	0.18	2.94	5.0 − 11.3	2.0 − 4.5
	0.204	0.0816	0.24	2.94	6.7 − 15	2.0 − 4.5
	0.204	0.0816	0.48	5.88	13.4 − 30	2.0 − 4.5
6-blade	0.102	0.034	0.10	2.94	2.8 − 6.3	2.0 − 4.5
	0.153	0.051	0.15	2.94	4.2 − 9.4	2.0 − 4.5
	0.204	0.068	0.20	2.94	5.6 − 12.5	2.0 − 4.5
	0.204	0.068	0.40	5.88	11.2 − 25	2.0 − 4.5

Power coefficients and tip-speed ratios

Figures 4.11 – 4.14 illustrate the effects of the key geometrical parameters, namely the number of blades, turbine radius, and chord length on both the optimal tip- speed ratio (TSR_{opt}) and the maximum power coefficient for three fixed solidity values.

The influence of the number of blades on the TSR_{opt} is highlighted in Figure 4.11. For all cases at a same radius (R = 2.9 m) but with varying the chord lengths and numbers of blades, the maximum performance occurs at the same TSR_{opt}. This observation indicates that TSR_{opt} is largely independent of both the blade chord length and number of blades. Similarly, for configurations with different radius (R = 2.9 m, 5.8 m, and $\sigma = 0.2$), the TSR curves closely overlap, suggesting that the turbine radius has negligible effect on the TSR_{opt}. Overall, these results demonstrate that the TSR_{opt} remains nearly constant despite variations in blade number, chord length, or turbine radius, and is primarily governed by turbine solidity. This finding underscores the importance of solidity as the dominant geometric parameter controlling the rotational speed at which maximum hydrodynamic efficiency is achieved.

The relationship between the number of blades and maximum turbine performance is observed to be inversely proportional, as shown in Figure 4.12. This effect is particularly pronounced at low solidity ($\sigma = 0.1$), where the three-blade configuration achieves a maximum power coefficient approximately 8% higher than that of the six-blade configuration. In contrast, for the higher solidity values, particularly for the commercial turbine scale ($\sigma = 0.2$, R = 5.8 m), the performance gain from reducing the number of blades is minimal, with only a 2% increase when transitioning from six to three blade configuration. Overall, turbine performance tends to decrease nearly linearly with an increasing number of blades. However, as the turbine scale increases, this dependency weakens, and the maximum power coefficient becomes nearly independent on the number of blades.

The relationship between the local Reynolds number and the maximum power coefficient depicted in Figure 4.13. For the low solidity ($\sigma = 0.1$, R = 2.9 m), turbine performance is highly sensitive to variations in the Re_c. In contrast, for the higher solidities, the maximum power coefficient becomes largely independent of Reynolds number. This trend is particularly evident for large turbine scale ($\sigma = 0.2$, R = 5.8 m), which often operate under field or open-water conditions, the Reynolds number impacts is neglected. Moreover, increasing solidity from $\sigma = 0.1$ to $\sigma = 0.15$ (at the same radius of 2.9 m) provides noticeable improvements in turbine performance, likely due to increased blade surface area and enhanced torque generation. However, further increasing solidity to $\sigma = 0.2$ results in only marginal gains. Those current observations suggest that turbine designs should aim for operating conditions in the Re_c-independent regime. Operating in this regime reduces sensitivity to Reynolds number effects, enhances performance, and simplifies the design and the optimization process. To better understand those observations, further

in-depth analyses and studies are required, which are proposed in the future work suggestions in Chapter 7.

In contrast to the nearly linear relationship between the blade number and the turbine performance, the blade chord length exhibits a non-linear effect with the maximum power coefficient, as shown in Figure 4.14. The chord length strongly influences local flow dynamics around the blades, making it a critical factor in determining the turbine's hydrodynamic behavior. This effect is further corroborated by the trends observed in both Figures 4.13 and 4.14, where variations in chord length produce similar impacts on turbine power as changes in the Reynolds number. From a design optimization perspective, to mitigate the performance losses associated with the low Reynolds number condition that could be improved by the geometrical parameter, such as the chord length, where the chord length should be large when the solidity and number of blades are predetermined. This highlights the chord length as a key geometric parameter that can be strategically adjusted to optimize turbine efficiency under specific operational conditions.

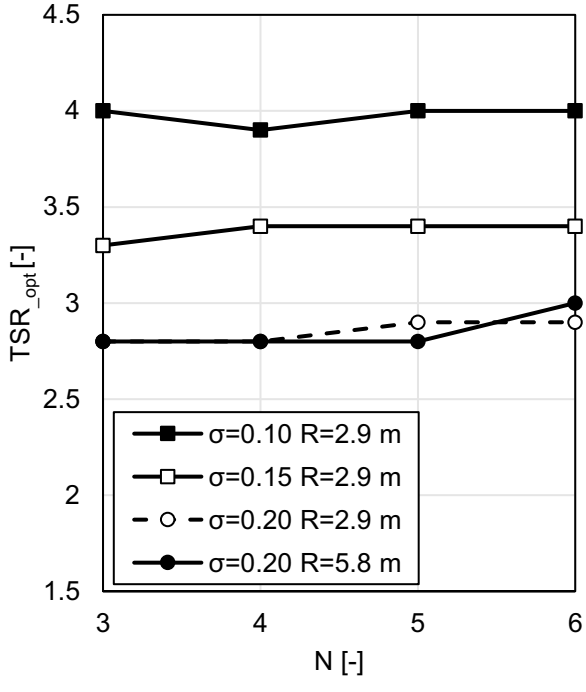


Figure 4.11. Effect on TSR_{opt} against number of blades. Adapted from Paper IV.

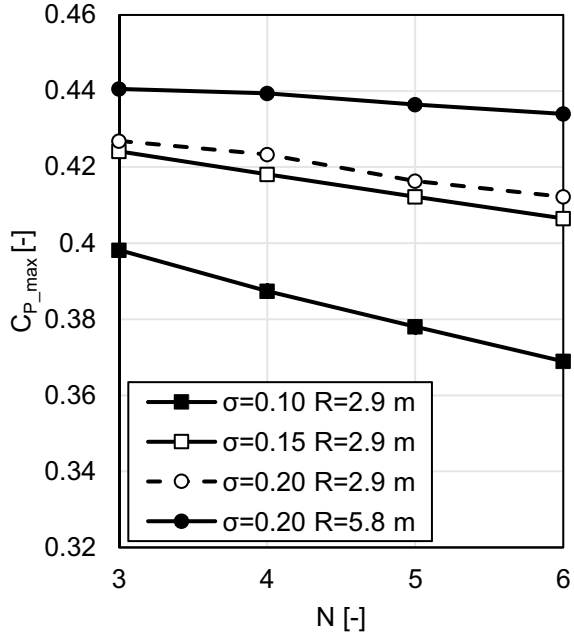


Figure 4.12. Effect on C_P against number of blades. Adapted from Paper IV.

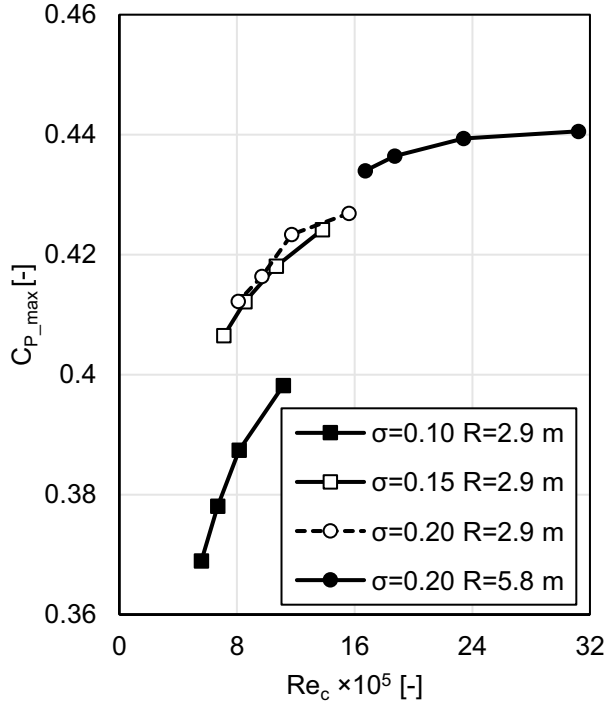


Figure 4.13. Effect on C_P against Reynolds number. Adapted from Paper IV.

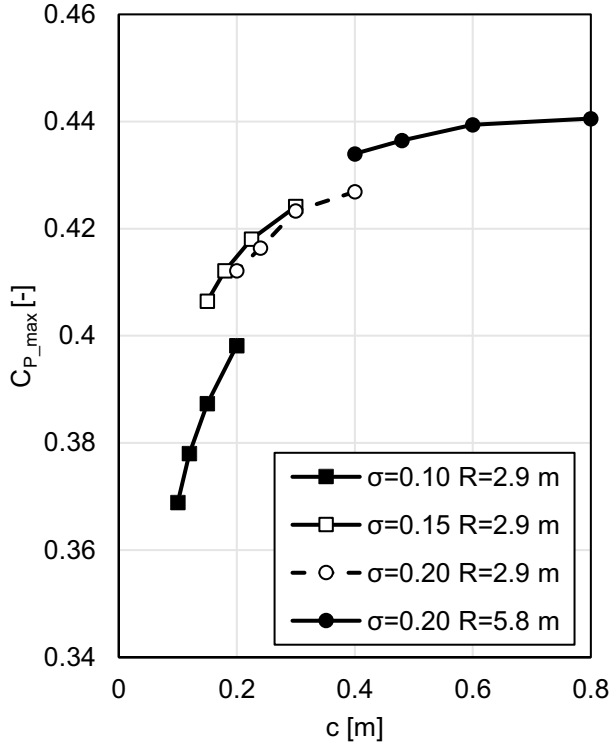


Figure 4.14. Effect on C_P against the chord length. Adapted from Paper IV.

Torque fluctuation

With regard to torque fluctuations, marine current turbines typically have relatively low moment of inertia compared to the inertia of the surrounding fluid. This characteristic enables the turbine to accelerate rapidly to its operational speed. For the test unit deployed in the river Dalälven, it is observed that the turbine can reach its peak speed within only a quarter of a revolution (Paper IV). Under such conditions, maintaining a uniform torque on the shaft is highly desirable, as it minimizes vibration, reduces mechanical stress, and a more even rotational speed. Figure 4.15 illustrates the impact of blade number on torque fluctuations for a commercial turbine scale (σ = 0.2, R = 5.8 m). Among the configurations analyzed, the three-bladed turbine exhibits the highest level of torque fluctuation across the entire range of tip-speed ratios (TSRs). Increasing the number of blades to four leads to a substantial reduction in torque oscillations, indicating a more uniform distribution of hydrodynamic loading throughout the rotational cycle. However, further increasing the blade count to five or six results in only small improvements in fluctuation reduction. Notably, configurations with four, five, and six blades exhibit comparable levels of torque smoothness.

From a design standpoint, these results indicate that while increasing blade number is an effective strategy for reducing torque fluctuations. A four-blade configuration appears to provide a favorable compromise between minimizing torque ripple and avoiding unnecessary increases in the number of blades, which could otherwise lead to such higher material costs of the turbine system.

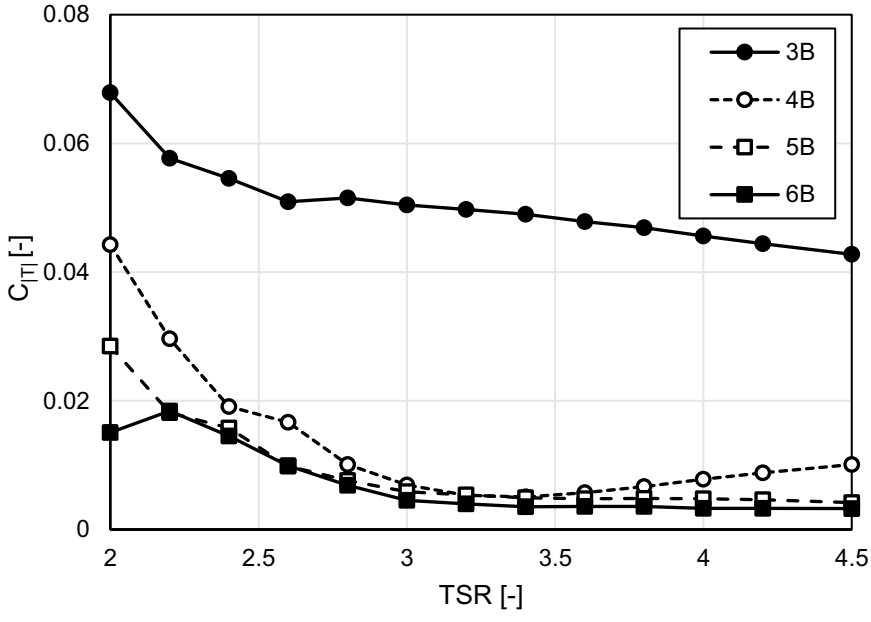


Figure 4.15. Torque amplitude coefficients of the upscaled turbine against TSRs at solidity $\sigma = 0.204$, R = 5.8 m. Adapted from Paper IV.

The two-dimensional numerical results for the power coefficient indicate that the three-bladed turbine configuration consistently achieves the highest performance across all examined solidity values. An inverse relationship is observed between the maximum power coefficient and the number of blades: turbines with fewer blades tend to produce higher C_P at a given solidity. This behavior can be attributed to the increase in chord length required to maintain constant solidity as the blade number decreases, which in turn raises the chord-based Reynolds number. The higher Re_c shifts the flow regime further into a fully turbulent state, improving lift generation and delaying flow separation, thereby enhancing turbine performance.

While configurations with fewer blades are advantageous in terms of power output, they are associated with increased torque fluctuations due to reduced blade-flow interaction frequency and less uniform load distribution over a rotation cycle. This effect is particularly critical in marine current applications, where unsteady loading can lead to increased structural fatigue and reduced operational lifespan.

From a design perspective, these findings highlight the need for a balanced optimization strategy. Although a lower number of blades (e.g., three blades) maximizes hydrodynamic efficiency, it may compromise load stability and structural reliability. Therefore, turbine design must consider both performance metrics and dynamic loading characteristics (such as torque ripple). The optimal configuration should be selected based on a trade-off between maximizing energy extraction and minimizing unsteady loads, taking into account the specific operational environment and structural constraints.

The numerical simulations conducted in this study are based on a two-dimensional approach, which inherently introduces several limitations when applied to VATs. In particular, important three-dimensional (3D) geometrical and flow features, such as blade finite span and supporting struts, are not represented in 2D models. One of the most significant omissions is the absence of blade tip effects, including the formation of tip vortices, which contribute substantially to aerodynamic losses in real turbines. As a result, 2D simulations generally underestimate total losses compared to more realistic 3D simulations performed. Despite these limitations, 2D simulations remain a valuable tool in the early stages of turbine design. Their significantly lower computational cost and reduced simulation time make them well-suited for preliminary analyses, sensitivity studies, and parametric optimization.

5. Conclusions

The thesis investigated the impact of several parameters on the performance and forces of the wind and marine current turbines using two-dimensional CFD models. Based on the outcome obtained from the study, the following conclusions can be made.

- The CFD models have been validated by the field experimental data, including measurements of the instantaneous turbine forces against the azimuthal angles, and the turbine power coefficients against a wide range of TSRs. The validation demonstrates the good agreement between the numerical simulations and experimental data. Furthermore, the CFD models have sufficient accuracy in capturing phenomena.
- The CFD numerical simulations can capture the phenomenon of the force drop influenced by the turbine shaft. This impact is visualized by decreases in the normal forces and turbine torque on the downstream side of both the wind and marine current turbines, contributing to the power losses.
- The numerical results demonstrate that the positive blade-pitch angles are more beneficial than the negative ones since they enable a greater area for extracting torque in both the upstream and downstream during rotations. The optimal pitch angle identified is +2°.
- For a given solidity, increasing the number of blades beyond three does not offer a benefit in the turbine performance. Among the configurations examined with 3, 4, 5, and 6 blades, the three-bladed configuration consistently demonstrates the highest power coefficient. However, in terms of the torque amplitude, increasing the number of blades generally provides clear advantages, as detailed in Paper IV. Specifically, increasing the number to 4 blades from 3 blades significantly reduces the torque fluctuations. Therefore, at the optimal design point, if minimizing torque amplitude is a critical criterion, the 4-blade configuration represents a more suitable choice than the 3-blade alternative.
- Reynolds numbers are sensitive to the turbine performance at the threshold of the laminar-to-turbulent transition. However, its impact becomes less significant or even negligible for the large-scale

turbine, particularly those at the commercial sizes. Therefore, during the design phase, it is essential to identify the minimum Reynolds number to ensure the turbine's operation beyond the sensitive regime and to mitigate any adverse effects associated with the low Reynolds number.

- The two- dimensional numerical models are facing limitations in accurately predicting turbine performance at low TSRs, primarily due to the exclusion of components such as struts and blade-tip effects. The impacts from the missing elements introduce greater fluctuations and instabilities in the simulation results at the low TSRs compared to the higher TSRs.

6. Summary of papers

This section presents a brief summary of the papers that are included in the thesis and the author's contribution.

Paper I

Fault analysis of a marine current vertical-axis turbine missing two blades

During the operation of a five-blade water turbine, two blades were found to be missing in the system, resulting in an asymmetric turbine. An investigation is carried out to evaluate the operation of both the original symmetric five-blade turbine and the asymmetric turbine with the missing blades. The differences between the two configurations, specifically in terms of the average rotational speed and normal force, are analysed and compared to the open-site river measurements in the free-spin condition (operating without loads). The experimental findings provide valuable insight for structural force design, demonstrating the asymmetric turbine configuration has extremely high forces acting on each remaining blade.

The author compiled the existing measurement data, wrote the majority of the text for the paper, and presented it orally at an international conference. She is the first author of the paper.

Paper II

Evaluation of the unsteady aerodynamic forces acting on a vertical-axis turbine by means of numerical simulations and open site experiments

Aerodynamic forces acting on the blades during the revolution of a straight-bladed vertical axis turbine are analysed. The turbine model is validated by the field experimental measurements that were conducted at Marsta, Sweden. The normal force is intensively investigated by comparing it to the experiment. Overall, a good agreement has been obtained between simulations and experiments. The obtained results are a valuable reference for the performance prediction since the lack of detailed experimental data often limits the reliability and effectiveness of numerical analyses.

The author performed the CFD simulations and wrote the majority of the text for the paper. Her contributions were in conceptualization, methodology,

software, validation, investigation, formal analysis, data curation, writing, reviewing, and editing. She is the first and corresponding author of the paper.

Paper III

Effect of pitch angle on power and hydrodynamics of a vertical axis turbine

This paper investigates the effect of blade-pitch angles on a vertical-axis water turbine with five straight blades. A wide range of pitch angles, from -5° to $+5^{\circ}$ was examined by using computational fluid dynamics (CFD). The CFD model was validated against experimental measurements that were performed at an open-site test location on the river Dalälven. The numerical results show a reasonably good agreement on the model validation, confirming the reliability of the model. Furthermore, the relevant physical phenomena were analysed, and the optimal pitch angle is identified, which has a 2.3% higher power coefficient than the reference case with zero pitch angle. This finding contributes valuable insight toward the design optimization, and understanding of the complex flow interaction in the vertical-axis turbines in marine current applications.

The author performed the CFD simulations and wrote the majority of the text for the paper. Her contributions were in conceptualization, methodology, software, validation, investigation, formal analysis, data curation, writing, reviewing, and editing. She is the first and corresponding author of the paper.

Paper IV

Assessing the influence of operational and geometric parameters on the hydrodynamic performance of vertical-axis marine current turbines

This study examined the influence of various parameters on the turbine performance and hydrodynamic forces of vertical-axis water turbines. A total of 16 turbine configurations were simulated by using two-dimensional computational fluid dynamics (CFD). The combined impacts of geometric parameters of blade number, turbine radius, chord length, and operational conditions of flow velocity, tip-speed ratio, and Reynolds number are systematically analysed. The intensive analysis of turbine performance, as well as the unsteady lift and drag forces are presented. One of the key findings is that the turbine performance can be enhanced by approximately 8% through optimization of the blade number at a given solidity.

The author performed the CFD simulations and wrote the majority of the text for the paper. Her contributions were in conceptualization, methodology, software, validation, investigation, formal analysis, data curation, writing, reviewing, and editing. She is the first and corresponding author of the paper.

Paper V

Integrated evaluation of Geometric and operating parameters on H-Darrieus hydrodynamic turbine performance and structural response

This paper studies the correlations of key geometric and operational parameters by using the Unsteady Reynolds-Averaged Navier Stokes (URANS) method. The numerical model has been well validated with the experimental data. The study outcome offers reliable insights into the hydrodynamic performance and operational stability through detailed analyses of the power coefficient and instantaneous hydrodynamic force curves, such as tangential, normal, axial, and lateral force components. The paper sheds some lights on the structural-force behaviours, addressing a gap in the literature where the revolution-resolve forces are seldom analysed in a few publications' studies

The author performed the CFD simulations in this paper. Her contributions were in conceptualization, validation, formal analysis, reviewing, and editing. She is the second author of the paper.

7. Future work suggestions

The work presented in the thesis could be continued in the following directions:

- Numerical simulations: Implementation in 3D

The CFD models in this thesis were in two-dimensional simulations, so the influence of the supporting struts and the full shaft was not taken into account. Particularly, the low TSR, the fluid dynamic reaction is much more complicated due to the unstable turbines. The 3D simulations might catch up to all phenomena occurring during revolutions.

- Impact of the turbine shaft

The impact of the turbine shaft on the force was noticed by the force-drop phenomenon. Its contribution should be investigated on hydrodynamics/aerodynamics and turbine performance as well. To have a solid conclusion on the shaft's impact, another study could be done on different values of the shaft radius.

- Impact of the chord-based Reynolds number

It has been observed that some turbine configurations were working in Reynolds-number-independent behaviour. It is necessary to simulate a larger number of different turbine configurations and operational working conditions to support conclusions such as how much it impacts the turbine performance.

- Some other interesting research topics

Such as the turbine supports, system influence, or multiple turbine system specially in the marine current area.

8. Sammanfattning på svenska

Det finns för närvarande två huvudtyper av kraftverk som används för att omvandla rörelseenergi från en fritt flödande fluid till roterande mekanisk energi: horisontalaxlade och vertikalaxlade kraftverk. För vertikalaxlade kraftverk är rotoraxeln vinkelrät mot den inkommande flödesriktningen, medan horisontalaxlade kraftverk har en rotationsaxel parallell med flödesriktningen. Horisontalaxlade kraftverk har vanligtvis generatorn monterad i ett maskinhus på toppen av ett torn, eller liknande stödstruktur, vilket ökar den strukturella belastningen, installationskostnaden och gör drift och underhåll mer komplicerat.

Vertikalaxlade kraftverk har flera fördelar jämfört med horisontalaxlade turbiner. En viktig fördel är att de inte behöver någon girmekanism för att orienteras mot flödet, vilket gör att de kan arbeta effektivt även vid varierande flödesriktningar. Dessutom kan tunga kritiska komponenter, såsom generatorn, placeras på marknivå för landbaserad vindkraft eller vid vattenytan för havsbaserad kraft. För strömkrafttillämpningar kan generatorn placeras nära botten, vilket ger en lägre tyngdpunkt och förenklar installation, drift och underhåll.

Dessa fördelar har bidragit till ett ökat forskningsintresse för vertikalaxlade kraftverk, särskilt på grund av deras enkla mekaniska design och möjligheten till lättare underhåll. Trots den mekaniska enkelheten är de strömningsmekaniska egenskaperna betydligt mer komplexa än för horisontalaxlade turbiner. Detta beror på att turbinbladen kontinuerligt upplever varierande anfallsvinkel under rotationen. Bladen utsätts för kraftigt tidsvarierande flöden, vilket ger upphov till flera utmanande fenomen, såsom dynamisk överstegring och interaktion med virvlar som avges från bladen under den främre halvan av varvet och sedan träffar bladen under den bakre halvan av turbinen. Dessa effekter gör det svårt att beräkna turbinens prestanda och därmed utmanande att optimera designen. Konsekvensen är att en djup förståelse av den underliggande strömningsmekaniken och dess koppling till strukturella effekter krävs för att utveckla effektiva och pålitliga vertikalaxlade kraftverk.

Vid Uppsala universitet har omfattande forskning bedrivits för att utveckla och analysera vertikalaxlade turbiner med raka blad. En viktig milstolpe var konstruktionen av ett vertikalaxlat kraftverk vid testanläggningen i Marsta 2006. Denna turbin hade tre raka blad. Som en fortsättning på forskningen utvecklades även ett vertikalaxlat kraftverk för strömkrafttillämpningar, vilket resulterade i en fembladig turbin som installerades i Dalälven. Dessa större

experimentanläggningar, som möjliggör studier av både atmosfäriska vindar och naturliga undervattensströmmar, erbjuder unika möjligheter att undersöka kraftverken under realistiska driftsförhållanden. Till skillnad från kontrollerade laboratoriemiljöer, såsom vindtunnlar eller släptankar, gör fältinstallationer det möjligt att studera effekterna av naturlig turbulens och variationer i den operativa miljön. Detta gör dem särskilt värdefulla för validering av teoretiska modeller och för att bedöma teknikens praktiska tillförlitlighet.

De experimentella mätningarna från dessa projekt har genererat stora mängder data, inklusive mätningar av effektfaktor och radiella bladbelastningar. Dessa data har använts för att fördjupa förståelsen av vertikalaxlade kraftverks prestanda, särskilt i relation till varierande belastning och dynamiskt svar. Tillgången till högkvalitativa fältmätningar har dessutom möjliggjort en grundläggande validering av de numeriska modeller som utvecklats i denna avhandling. Kombinationen av experimentella observationer och strömningsmekaniska simuleringar ger ett robust ramverk för att förbättra beräkningsprecisionen och för att vidareutveckla turbindesignen, vilket i sin tur ökar kraftverkens effektivitet och pålitlighet.

För vindenergitillämpningar har fokus legat på en numerisk analys av de varierande aerodynamiska krafterna som verkar på turbinen. Målet har varit att förbättra förståelsen av turbinens strukturella svar, inklusive lastkaraktäristik och dess påverkan på konstruktionen. För strömkrafttillämpningar har studien istället fokuserat på hur geometriska parametrar, såsom antal blad, turbinradie och bladens vinkel, samt driftförhållanden, såsom löptal och flödeshastighet, påverkar belastningen på turbinen.

Den metod som använts är beräkningsströmningsmekanik (CFD). Arbetet omfattar följande steg: först definieras geometrin för simuleringsdomänen, därefter genereras beräkningsnätet, randvillkoren specificeras och slutligen konfigureras den numeriska lösaren. För att säkerställa pålitliga och högupplösta resultat har modellerna först verifierats genom studier av rutnätsoberoende och tidsstegsoberoende. Denna verifieringsprocess säkerställer att lösningen inte påverkas av numeriska fel. Modellen har därefter validerats mot experimentella data, vilket säkerställer att den teoretiska modellen överensstämmer med verkligheten. Tillsammans säkerställer verifiering och validering att de numeriska resultaten är både tillförlitliga och korrekta.

Resultaten från studierna ger detaljerade insikter i hur både designparametrar och driftförhållanden påverkar den genererade effekten samt hur de dynamiska krafterna varierar. Dessa resultat utgör en robust grund för optimering av vertikalaxlade kraftverk och kan tillämpas inom både vindkraft och marin strömkraft.

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Inside the academic world

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